

# Zillow House Prices

**Team:** Julio Luque and Kevin Somu

## **Problem Statement:**

In today's competitive real estate environment, accurate forecasting of real estate prices is critical to homebuyers, sellers, investors, and financial institutions. The traditional methods of valuation: Comparative Market Analysis and appraisals, are usually based on history of past sales and personal opinion, leading to unpredictable and sometimes inaccurate pricing. This unpredictability is a big issue for homebuyers in search of best value houses and investors seeking maximum return on investment.

The rapid growth of online portals, such as Zillow, has revolutionized real estate through the provision of rich data regarding property characteristics, price patterns, and neighborhood metrics. The challenge in deriving insightful information and uncovering undervalued properties comes from the nature of high-dimensionality and complexity of such information. The current predictive models, such as Zillow's Zestimate, overestimate or underestimate house prices due to limited feature selection and a lack of model interpretability (Zillow, 2024).

This project attempts to bridge this void by designing a machine learning forecasting model based on Zillow's API for predicting house selling prices and identifying best-value houses in given locations. Our model is designed to predict the final selling price of a house, rather than its listed or appraised value.

Moreover, the model will enable search by location, enabling location-based analysis of real estate markets and assisting in finding best value properties for buyers. This kind of approach not only enhances predictability but also promotes transparency and interpretability and corrects the limitation of existing methods for real estate valuation.

## **Data Source:**

For this project, we accessed our real estate data dynamically via the Zillow API on RapidAPI. Rather than using fixed datasets, we built a full end-to-end automated data intake pipeline that programmatically extracts property-level data for any given U.S. city and state combination. This dynamic methodology makes it simple for us to easily have our model trained on various geographic markets without recoding or manually harvesting data. To gather property data, our pipeline first queries Zillow's "propertyExtendedSearch" endpoint to retrieve a list of properties that are either "For Sale" or "Recently Sold" in a specified city and state. This endpoint provides basic metadata, including the unique property ID (zpid) for each listing. We then use the zpid to make a second, more detailed call to Zillow's "property" endpoint, which returns comprehensive property-level information such as price history, school ratings, garage capacity, pool availability, and nearby comparable homes. This two-step process allows us to efficiently collect both broad and deep data needed for accurate modeling.

Our pipeline uses two primary Zillow API endpoints: one for "For Sale" properties and one for "Recently Sold" properties. When a new city-state combination is executed in the model, two new CSV files will be created (City\_State\_ForSale, and City\_State\_RecentlySold). To avoid using too many API calls and to speed up the model, when re-running the code, if the files for the city-state are saved in the same working directory, it will run using the both CSVs instead of the API. We then merge both sets into a single combined set, and this merged set is used to train and test the model via cross-validation. As a means of correcting for the unequal reliability of the price labels, we place more emphasis on "Recently Sold" data (weight = 1.0) and less emphasis on "For Sale" data (weight = 0.5). This weighted approach enables the model to learn from both actual market sales and current listing trends. We then cross-validate on this combined dataset maintaining the same weighting approach for each fold to assess the predictive capability of the model.

## **Data Preprocessing & Feature Engineering:**

After retrieving raw property listings from the Zillow API, our pipeline conducts significant data preprocessing and feature engineering to prepare the dataset for modeling. These processes are essential to enhancing data quality, reducing noise, and achieving consistency across city-level models.

We began by dropping homes priced below \$30,000 to remove distressed or incomplete listings that could skew the model. We dropped home types that were classified as manufactured, empty lots, and unknown classifications. Entirely missing columns or non-numeric identifiers like zipid, address, and longitude were dropped to lower the feature space. In addition, homes that were missing a price field were excluded from the dataset.

Next, we engineered additional features to augment the dataset. Two new variables measured school quality: the highest-rated proximate school (school\_max\_rating) and the count of schools listed for the property (school\_count). We also computed house\_age as the year today minus the year in which the house was built, which is a good approximation of property condition and desirability.

To approximate local market conditions, we examined each property's nearbyHomes list and searched for homes of the same type of home. For each of these nearby homes, we extracted its price and living area and computed its price per square foot. We then created separate lists for recently sold and for-sale properties. From the two lists, we calculated NearbyAvgPrice by calculating a weighted average of house prices, assigning full weight (1.0) to RecentlySold properties and partial weight (0.5) to ForSale properties. This weighting reflects the higher reliability of actual transaction prices relative to asking prices. The same calculation was performed to compute NearbyPricePerSqFt, using the same weighted averaging method for pricing-per-square-foot figures. These fields were not included in the model, but they were used to calculate the NearbyScaledPrices field. This is the NearbyPricePerSqFt field times the living

area of the house except when the `NearbyPricePerSqFt` is greater than an outlier threshold of \$10,000, and then we use the `NearbyAvgPrice` instead.

Imputation and preprocessing were carried out such that there is no data leakage. Missing values in `lotAreaValue` were imputed by using `livingArea` as a fill-in value. One-hot encoding was applied on categorical features such as `homeType` and `furnished`, and numerical features were scaled using `StandardScaler`.

To guard against outliers that would distort the regression model, we used a Random Forest Regressor to predict home prices and calculate residual errors. Listings whose residuals fell beyond a given limit were removed from the training set, effectively pruning outlier listings.

## Methodology:

To evaluate our approach for predicting housing prices, we developed a modeling pipeline that compares the performance of four different regression algorithms: Multiple Linear Regression (MLR), Random Forest, Gradient Boosting, and Neural Network. These models are tested across three distinct geographic regions within the Miami metropolitan area, two suburbs (e.g., Doral and Kendall) and one central urban area (Miami proper) to assess how well each model generalizes across different housing markets.

In addition to evaluating the performance of the above models, the code outputs a dataset that ranks properties from most undervalued to least, based on the difference between the model's predicted price and the listed price in the data. This can help potential buyers in identifying properties that are priced below their estimated market value.

To begin, we focus our exploratory data analysis (EDA) and modeling efforts on the city of Doral. By conducting EDA on all four models in this initial location, we aim to gain a deeper understanding of the dataset's structure, potential challenges, and early indicators of model behavior before applying the full pipeline to the remaining regions.

## Exploratory Data Analysis

As part of our exploratory data analysis for our Multiple Linear Regression (MLR) model, we tested the residuals to check the fit of the model. In Figure 1, the residual plot against predicted values shows that residuals hover around zero for most of the predicted values, showing a good linear fit for houses in the middle range. We do notice increasing variance in residuals as the predicted prices get higher, and there are some high-impact outliers in both positive and negative directions. This "fanning out" pattern suggests heteroskedasticity, which means the model performs less reliably on more costly properties, likely due to non-linear relationships not included in the MLR.

Also, in Figure 1, we plotted the distribution of residuals as a histogram with kernel density estimate. It is zero-centered and approximately bell-shaped like a normal distribution. This looks

like the MLR model produces good predictions on average because most of the residuals are clustered around zero. The residuals distribution suggests the errors are randomly distributed, which goes with principal linear regression assumptions and provides more confidence in the model's performance for properties.

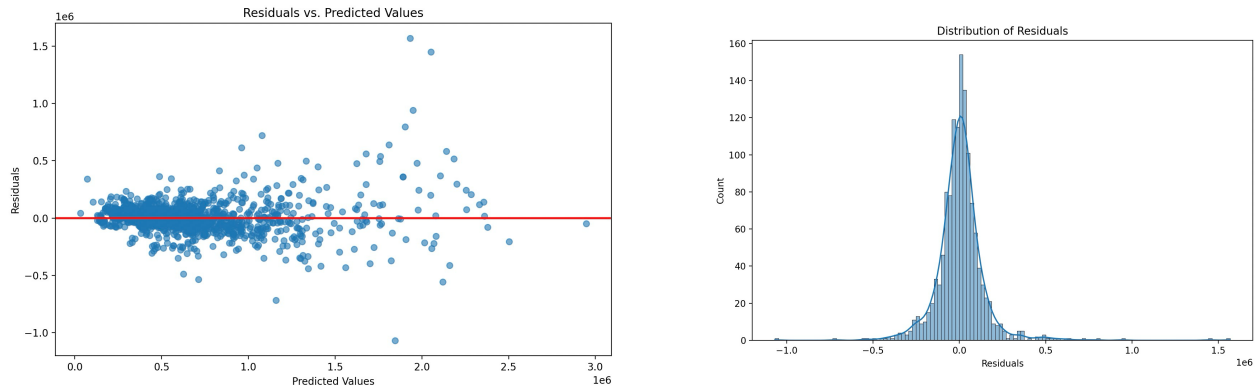


Figure 1: Residual Analysis for MLR

The residual plot of the Neural Network model demonstrates a pattern very similar to that of the MLR model. In the residual vs. fitted values plot, residuals are still centered around zero for most mid-range predictions but are dispersed for predicted prices at higher ranges, indicating residual heteroskedasticity. The residual distribution is also heavily concentrated around zero and is somewhat right-skewed, very similar to the residual histogram of MLR. These similarities indicate that, even with the Neural Network's ability to model non-linearly, it does not outperform MLR to any great extent in terms of capturing pricing complexities in this dataset. Especially for higher-priced houses where both models have larger, more volatile errors. Because residual patterns for the Random Forest and Gradient Boosting models were also extremely similar, we turn our attention to feature importance analysis to gain a more nuanced understanding of how these ensemble methods weight predictors. Readers can refer to the A2, A4, A7, A8, A11, and A12 in the Appendix for residual plots for these models.

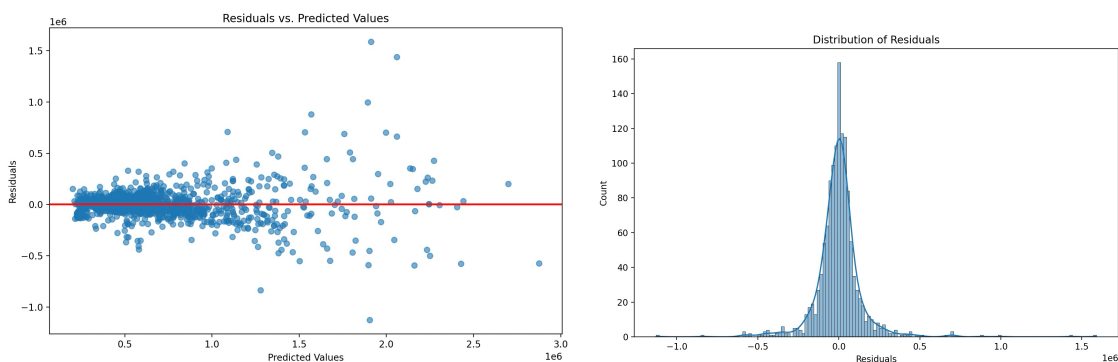




Figure 2: Residual Analysis for Neural Network

To better understand behavior in the Random Forest model, we examined the relative importance of all input features in figure 3. The feature importance plot from this analysis shows a very strong predominance by NearbyScaledPrices, accounting for over 85% of the model's decision-making ability. This indicates that the model is deriving very significant leverage from local market context, specifically price signals from proximate homes to make its predictions. All the other attributes, including lotAreaValue, livingArea, monthlyHoaFee, and structural attributes like bathrooms and bedrooms, are only contributing to small proportions relative to the others. This lack of balanced distribution of importance suggests that, in the case of the Doral market, spatial context of prices may be more informative than specific property attributes in estimating home values.

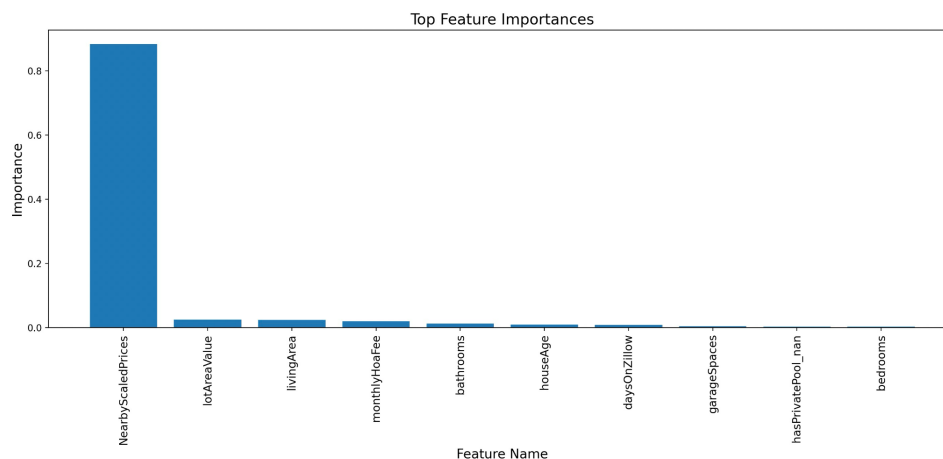


Figure 3: Feature Importance for Random Forest

To supplement the global feature importance analysis, we calculated SHAP (SHapley Additive exPlanations) values to better understand how individual features affect predictions across homes in Figure 4. Consistent with the Random Forest feature importance plot, NearbyScaledPrices is the most significant predictor, with a strong positive influence on model output when the value is high. The SHAP summary plot also reveals secondary contributors including lotAreaValue, houseAge, and livingArea, with more subtle and sometimes bidirectional influences on predicted prices. Notably, features like zipcode, monthlyHoaFee, and even categorical features like homeType and propertyCondition have some but weak influence, which suggests that while the model is dominated by spatial price signals, it does contain other contextual and structural information in some cases. EDA of the Gradient Boosting model also revealed very similar findings, and respective residual plots and SHAP visualizations are included in the A4, A8, and A12 in the Appendix. EDA for all 4 models was done on the Miami and Kendall markets as well, which you can review in the A5-A12 in the Appendix.

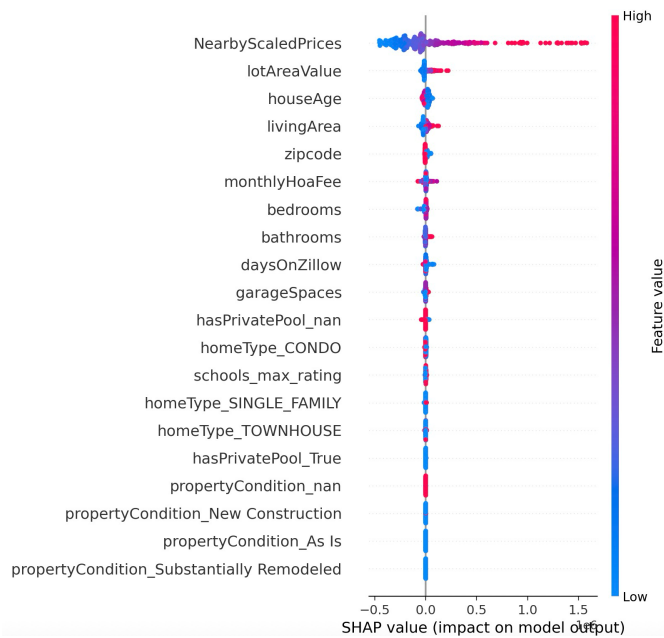


Figure 4: SHAP Values for Random Forest

## Modeling

For the MLR model, we used a Weighted Least Squares (WLS) approach from a combined dataset of "Recently Sold" and "For Sale" units. MLR was selected as a baseline model for its simplicity, transparency, and interpretability. It allows us to understand how each feature contributes on its own to predict the price and provides a valuable point of reference for comparing more complex models. We weighted "Recently Sold" listings by 1.0 and "For Sale" listings by 0.5 to reflect the higher reliability of completed sale data. We evaluated the model using 5-fold cross-validation by  $R^2$  and Mean Absolute Error (MAE) metrics. After training on the full dataset, we used the model to forecast prices for homes that are currently available and computed an "Undervalued Score" to compare with actual list price.

The Neural Network model was included to capture any potential non-linear interactions and feature interactions that MLR might not detect. We employed a multi-layer perceptron with three hidden layers (128, 64, and 32 neurons) and ReLU activation and Adam optimizer. Although more difficult to read than linear models, neural networks can be capable of capturing complex high-dimensional relationships between data, useful in real estate where price depends on nuanced combinations of variables. The same weighted training strategy and cross-validation technique were employed. The network was utilized upon validation to predict prices and identify undervalued properties from listing discrepancies.

For the Random Forest model, we trained an ensemble of 100 decision trees with depth control and minimum leaf size to avoid overfitting. Random Forests are effective with tabular data and provide good performance with minimal parameter tuning. The ability to handle non-linearities and feature interactions made it a suitable option for us. They also include in-built feature importance scores, giving us some interpretability and improved accuracy over simple models.

Following cross-validation, the model is used to generate undervaluation insights for active listings.

Finally, we used Gradient Boosting via XGBoost (XGBRegressor) as a high-speed ensemble learning algorithm that is famous for its improved accuracy on structured data. Gradient Boosting builds trees in sequence so that it can focus on correcting errors committed by the preceding trees. We used 1,000 estimators with learning rate 0.05 and max depth 10. To prevent leakage, 30% of the "For Sale" postings were set aside as a test set, and the remaining 70% were used for training along with the full "Recently Sold" data. Gradient Boosting was chosen because of its ability to deliver high accuracy and also offers tools like SHAP for post-hoc interpretability. We trained and cross-validated the model and then employed it to predict on held-out listings in order to estimate predicted prices and undervaluation scores.

**Evaluation and Results:**

We tested our four models' performance on three South Florida real estate markets. We began with Doral, a suburb, as our initial test case. We then conducted testing in Miami, a more complex and larger urban market. As initial evidence revealed the models performing stronger in Doral than Miami, we added Kendall, a second suburban city, into our research to validate if Doral's superior performance was due to its smaller market size. Finally, we investigated the relative model performances in each of the three markets to determine which models predict prices most accurately

|                   | R <sup>2</sup> | MAE        |
|-------------------|----------------|------------|
| MLR               | 0.8733         | \$92046.69 |
| Neural Network    | 0.8347         | \$95546.81 |
| Random Forest     | 0.8993         | \$72896.75 |
| Gradient Boosting | 0.9012         | \$71676.29 |

Table 1: Doral Average Cross Validation Results

All four models performed quite well in the Doral market with high R<sup>2</sup> values indicating good overall predictive capability. The best performing model was Gradient Boosting with the highest R<sup>2</sup> 0.9012 and lowest MAE \$71,676 and therefore the most accurate model for the suburban area. Surprisingly, the Neural Network model actually did worse on both metrics, producing the worst R<sup>2</sup> 0.8347 and highest MAE \$95,546, and possibly suggesting it is not calibrated to the structure of this data. While the baseline model was a MLR model, it was nevertheless competitive with an R<sup>2</sup> value of 0.8733, but a higher MAE (\$92,046), implying that ensemble methods are better at capturing the Doral market complexities.

|                   | R <sup>2</sup> | MAE          |
|-------------------|----------------|--------------|
| MLR               | 0.6285         | \$1004409.46 |
| Neural Network    | 0.704          | \$964931.54  |
| Random Forest     | 0.6004         | \$622942.77  |
| Gradient Boosting | 0.5615         | \$559170.37  |

Table 2: Miami Average Cross Validation Results

In comparison with Doral, the models performed much poorer in the Miami market with significantly lower R<sup>2</sup> values. This signifies greater variability and complexity in the city's housing data as a result of a wider range of prices. The Neural Network model posted the best R<sup>2</sup> at 0.704, which indicates it performed best in identifying patterns in this complex market. All the models, however, had extremely high MAE, with the errors reaching or exceeding \$1 million since they were representing huge price gaps between luxury and affordable properties in Miami. It was surprising that although Gradient Boosting had the lowest MAE of \$559,170, its R<sup>2</sup> (0.5615) was the lowest, which means that although it performed better at predicting average prices, it performed worse at explaining all variance. In conclusion, the research brings out the challenge of property price modeling in big, diverse urban markets like Miami, where non-linear models could partially correct but be bounded.

|                   | R <sup>2</sup> | MAE         |
|-------------------|----------------|-------------|
| MLR               | 0.8855         | \$144821.29 |
| Neural Network    | 0.8804         | \$145230.90 |
| Random Forest     | 0.8903         | \$120482.72 |
| Gradient Boosting | 0.892          | \$115777.76 |

Table 3: Kendall Average Cross Validation Results

Given superior performance of all the models in Doral compared to Miami, we tried to ascertain whether smaller and more homogeneous markets tend to have better predictive results. To test this, we tested the models in Kendall, another suburban town in the Miami metropolitan area. Like Doral, Kendall also consistently had high R<sup>2</sup> values in all the models, with Gradient Boosting again performing the best R<sup>2</sup> = 0.892 and lowest MAE \$115,778. Random Forest followed closely, with MLR and the Neural Network also performing well with very small differences in accuracy. However, despite the high R<sup>2</sup> values, MAE values were considerably higher in Kendall than in Doral, suggesting that prediction errors can vary across smaller

markets. These results lend credence to the argument that ensemble models are good generalizers in suburban markets, although some variability in error magnitudes continues to exist.

| City    | Models            | R <sup>2</sup> | MAE         |
|---------|-------------------|----------------|-------------|
| Doral   | Gradient Boosting | 0.9012         | \$71676.29  |
| Kendall | Gradient Boosting | 0.892          | \$115777.76 |
| Miami   | Gradient Boosting | 0.5615         | \$559170.37 |
| Miami   | Neural Network    | 0.704          | \$964931.54 |

Table 4: South Florida Models with Best Results

In all three markets, Gradient Boosting was the most stable and accurate model, particularly in the suburban areas of Doral and Kendall, where it generated the highest R<sup>2</sup> values (0.9012 and 0.892, respectively) and the lowest MAE values. These results suggest that Gradient Boosting is best suited for structured, less volatile housing markets. On the other hand, Miami was harder, with all models less accurate and with significantly higher MAE. Surprisingly, while Gradient Boosting had the lowest value for MAE in Miami, its R<sup>2</sup> was the worst, indicating poor explanatory power despite the average predictions being decent. Interestingly, the Neural Network exhibited relatively strong R<sup>2</sup> performance in Miami 0.704 even with high MAE, which implies that neural models can perhaps better capture the complexity and nonlinear behavior of bigger city markets. Overall, these results attest to the importance of market context in selecting and evaluating predictive models.

## Excel Output

In addition to evaluating model performance across different markets, our project was designed for actual use cases. Each model has its own standalone .py script to implement, and upon execution, produces a CSV output (City\_State\_ranked) of a ranked list of properties by "Undervalued Score" - the predicted price minus current listing price. This CSV is created on the initial execution of a model for a particular market. After running another model for the same market afterward, the CSV will be updated with new predictions and re-ranks the properties automatically. This feature makes our research work a valuable application that can guide homebuyers, investors, and analysts to identify potentially undervalued homes in a location. While our primary analysis was focused on comparing model performance for suburban vs. urban markets, this capability makes the codebase applicable in settings beyond the academic environment to inform real estate decisions.

## Conclusion:

In conclusion, our research showed that predictive model performance relies heavily on the type of real estate market. Ensemble models, particularly Gradient Boosting, tended to perform superior to the other models in suburban areas like Kendall and Doral, where market trends are bound to be more stable. In comparison, model accuracy shrunk in the more complex Miami market, in which the Neural Network recorded a high  $R^2$  with vast prediction error, suggesting potential for deep learning models to uncover patterns in high-variance metropolitan settings. As future areas of study, deeper, more intricate deep learning architectures could be explored specifically in large, complex cities where non-linear patterns prevail. Additionally, extending this work beyond Florida and using it to apply to suburbs in other major cities could help verify if small markets consistently allow for improved predictive performance across different regions of the country. Not only do these findings highlight the importance of market segmentation in model development but also open up new opportunities for optimizing real estate pricing tools on a national scale. To support real world application, our code exports a CSV file, which is sorted by undervaluation score based on the model that is run. The user can input any city-state combination in the United States, and the model will output the latest data into a CSV file. This file is updated each time a new model is run, making the tool valuable to future users.

## References:

1. *What is a zestimate? Zillow's zestimate accuracy.* Zillow. (2024, September 12). <https://www.zillow.com/z/zestimate/>
2. *Apimaker. (n.d.). Zillow.com API. RapidAPI.* <https://rapidapi.com/apimaker/api/Zillow.com>

## Appendix (Model Outputs):

### A1: Doral, FL (MLR)

Cross-Validation R<sup>2</sup> Scores: [0.8705926027925746, 0.8503896562847173, 0.9233527004619619, 0.8498238218055744, 0.8724221840357018]  
Average R<sup>2</sup> Score: 0.8733

Cross-Validation MAE Scores: [88926.07412021619, 89208.94945396412, 86854.0032909514, 95801.84024815448, 99442.59845583155]  
Average MAE Score: 92046.69

#### Model Summary:

##### WLS Regression Results

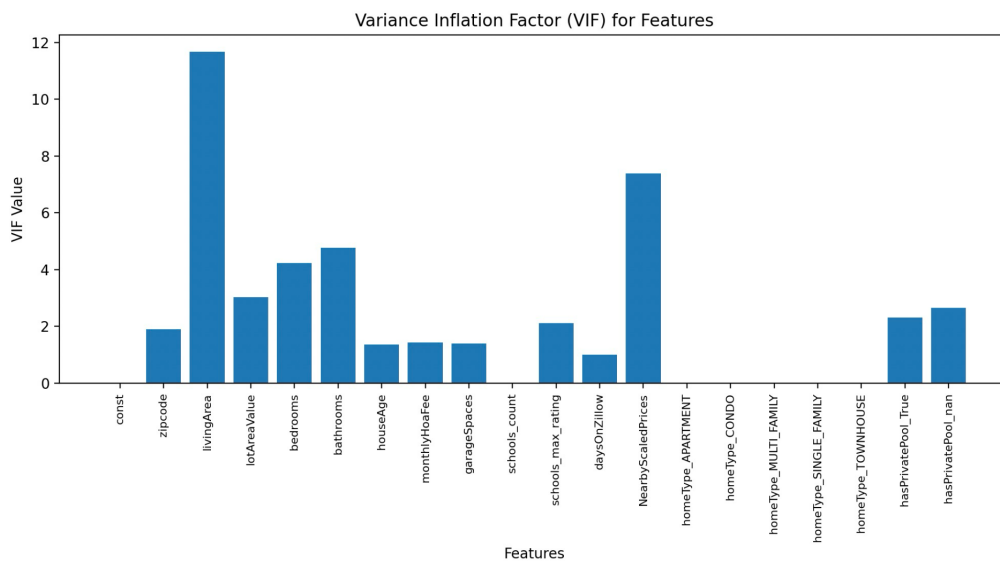
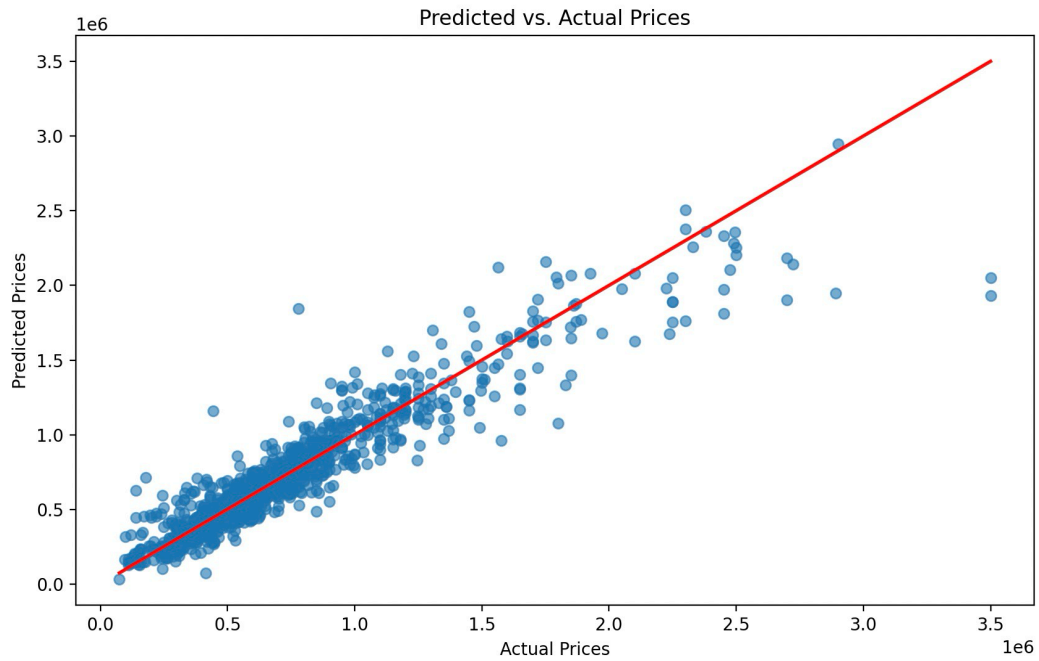
```
=====
=====
Dep. Variable:          y  R-squared:          0.878
Model:                  WLS  Adj. R-squared:    0.877
Method:                 Least Squares  F-statistic:    548.3
Date:                   Sat, 19 Apr 2025  Prob (F-statistic):    0.00
Time:                   16:20:29  Log-Likelihood:    -17447.
No. Observations:       1308  AIC:              3.493e+04
Df Residuals:           1290  BIC:              3.502e+04
Df Model:                17
Covariance Type:        nonrobust
=====
=====

```

|            | coef       | std err  | t      | P> t  | [0.025    | 0.975]   |
|------------|------------|----------|--------|-------|-----------|----------|
| const      | 7.47e+05   | 2.43e+04 | 30.717 | 0.000 | 6.99e+05  | 7.95e+05 |
| zipcode    | -1.017e+04 | 5379.620 | -1.890 | 0.059 | -2.07e+04 | 383.921  |
| livingArea | -4110.0325 | 2.37e+04 | -0.173 | 0.862 | -5.06e+04 | 4.24e+04 |

|                        |            |                   |           |       |           |           |
|------------------------|------------|-------------------|-----------|-------|-----------|-----------|
| lotAreaValue           | 2.675e+04  | 7603.267          | 3.518     | 0.000 | 1.18e+04  | 4.17e+04  |
| bedrooms               | 1.641e+04  | 8291.121          | 1.979     | 0.048 | 141.745   | 3.27e+04  |
| bathrooms              | 4.464e+04  | 8841.105          | 5.049     | 0.000 | 2.73e+04  | 6.2e+04   |
| houseAge               | -1.476e+04 | 4747.582          | -3.109    | 0.002 | -2.41e+04 | -5447.273 |
| monthlyHoaFee          | -7885.9828 | 4589.177          | -1.718    | 0.086 | -1.69e+04 | 1117.087  |
| garageSpaces           | 9690.4038  | 4923.493          | 1.968     | 0.049 | 31.472    | 1.93e+04  |
| schools_count          | 5.331e-11  | 1.35e-11          | 3.961     | 0.000 | 2.69e-11  | 7.97e-11  |
| schools_max_rating     | 2.559e+04  | 5853.316          | 4.373     | 0.000 | 1.41e+04  | 3.71e+04  |
| daysOnZillow           | -952.8739  | 1448.524          | -0.658    | 0.511 | -3794.596 | 1888.848  |
| NearbyScaledPrices     | 4.596e+05  | 1.77e+04          | 25.930    | 0.000 | 4.25e+05  | 4.94e+05  |
| homeType_APARTMENT     | 6.095e+04  | 3.59e+04          | 1.696     | 0.090 | -9554.975 | 1.31e+05  |
| homeType_CONDO         | 1.792e+05  | 1.52e+04          | 11.824    | 0.000 | 1.49e+05  | 2.09e+05  |
| homeType_MULTI_FAMILY  | 1.002e+05  | 4.98e+04          | 2.014     | 0.044 | 2607.597  | 1.98e+05  |
| homeType_SINGLE_FAMILY | 2.378e+05  | 1.77e+04          | 13.445    | 0.000 | 2.03e+05  | 2.73e+05  |
| homeType_TOWNHOUSE     | 1.688e+05  | 1.5e+04           | 11.289    | 0.000 | 1.39e+05  | 1.98e+05  |
| hasPrivatePool_True    | -1.199e+05 | 2.85e+04          | -4.212    | 0.000 | -1.76e+05 | -6.4e+04  |
| hasPrivatePool_nan     | -2.302e+05 | 2.61e+04          | -8.826    | 0.000 | -2.81e+05 | -1.79e+05 |
| =====                  |            |                   |           |       |           |           |
| =====                  |            |                   |           |       |           |           |
| Omnibus:               | 657.789    | Durbin-Watson:    | 1.889     |       |           |           |
| Prob(Omnibus):         | 0.000      | Jarque-Bera (JB): | 36772.665 |       |           |           |
| Skew:                  | 1.544      | Prob(JB):         | 0.00      |       |           |           |
| Kurtosis:              | 28.791     | Cond. No.         | 1.04e+17  |       |           |           |
| =====                  |            |                   |           |       |           |           |
| =====                  |            |                   |           |       |           |           |





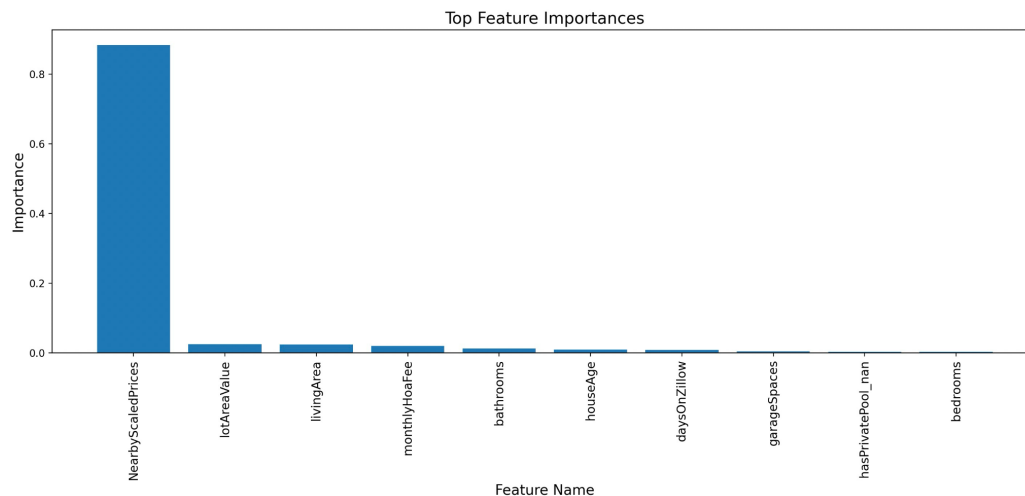
## A2: Doral, FL (Random Forest)

Cross-Validation R<sup>2</sup> Scores: [0.9160594531469071, 0.9105001771639171, 0.9116481588105216, 0.8941421714287375, 0.8640882326963113]  
Average R<sup>2</sup> Score: 0.8993

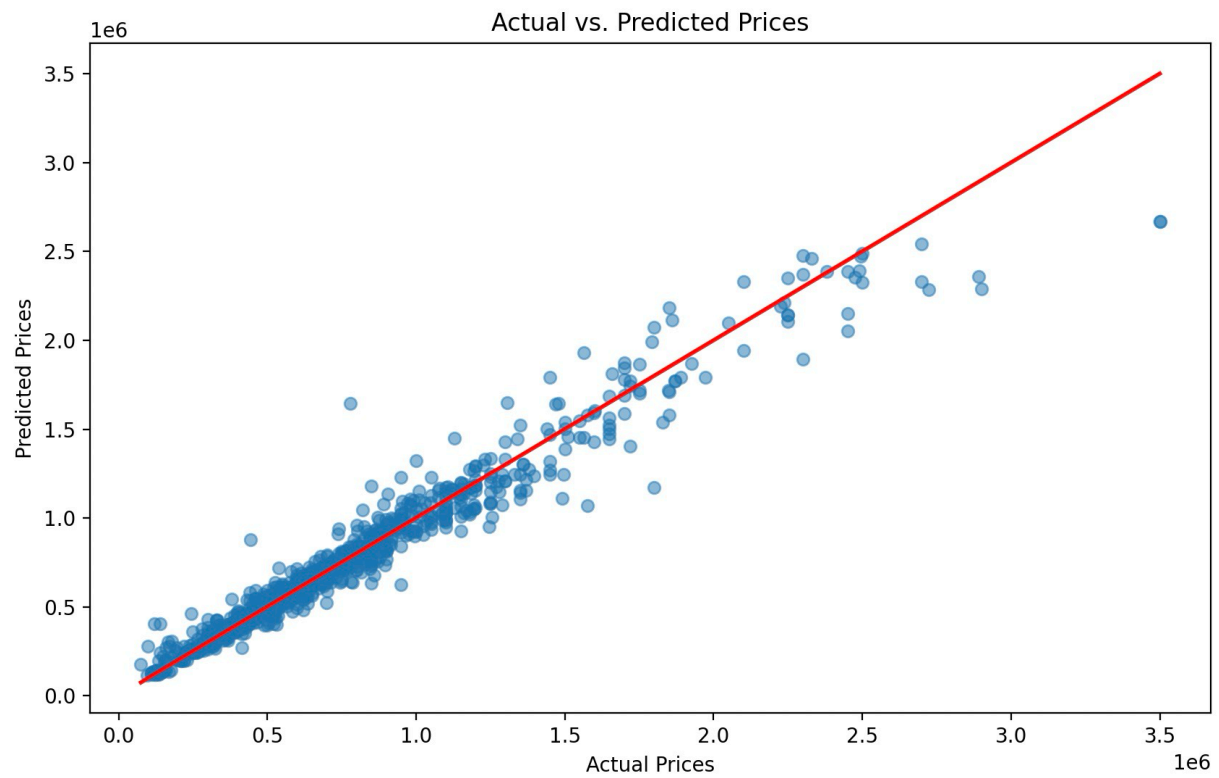
Cross-Validation MAE Scores: [67341.98024771466, 65894.4054201718, 74645.07894274211, 72944.0317016523, 83658.23810728274]  
Average MAE Score: 72896.75

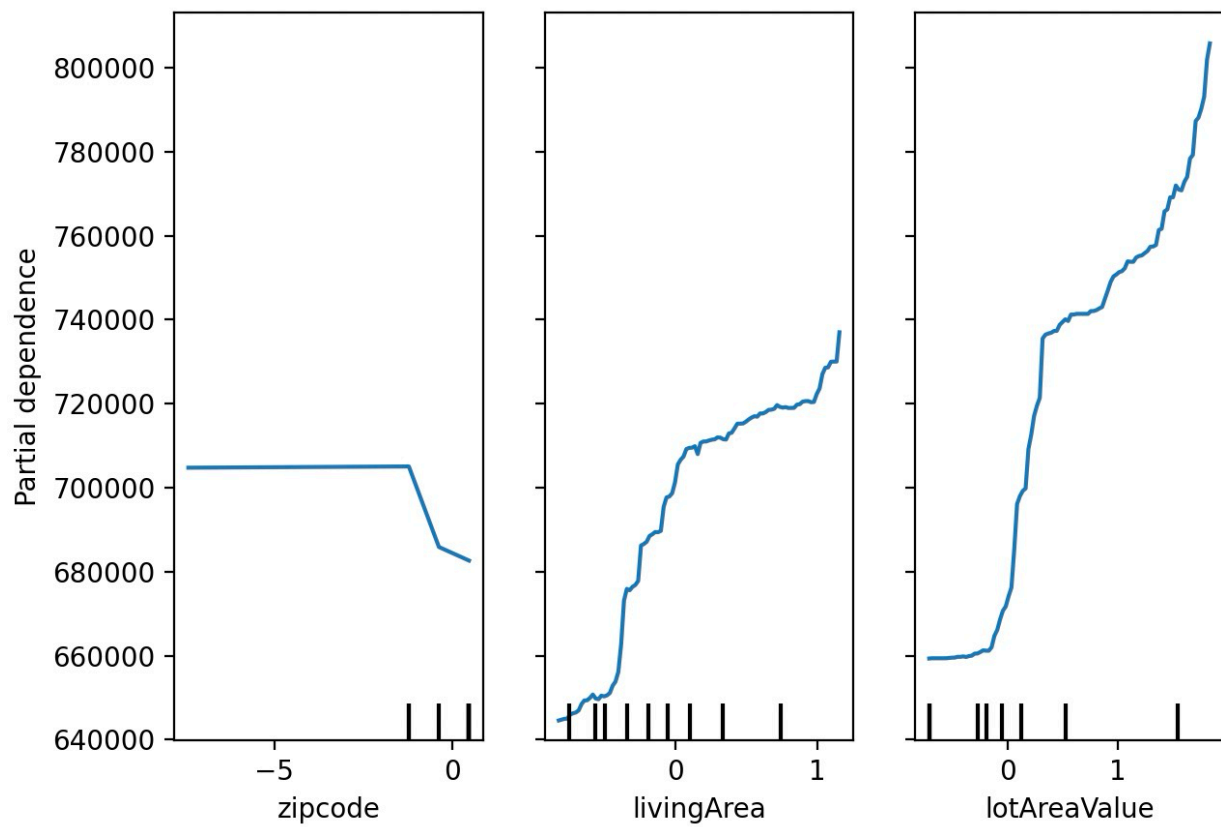
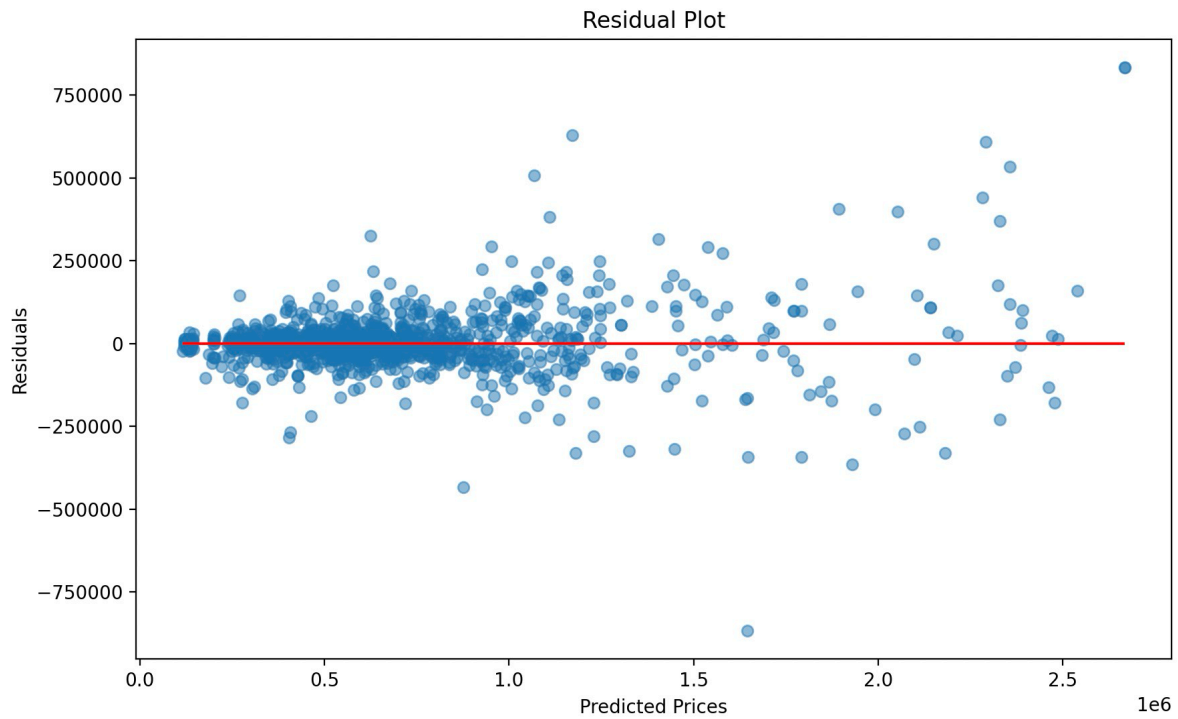
Top 10 Most Important Features:

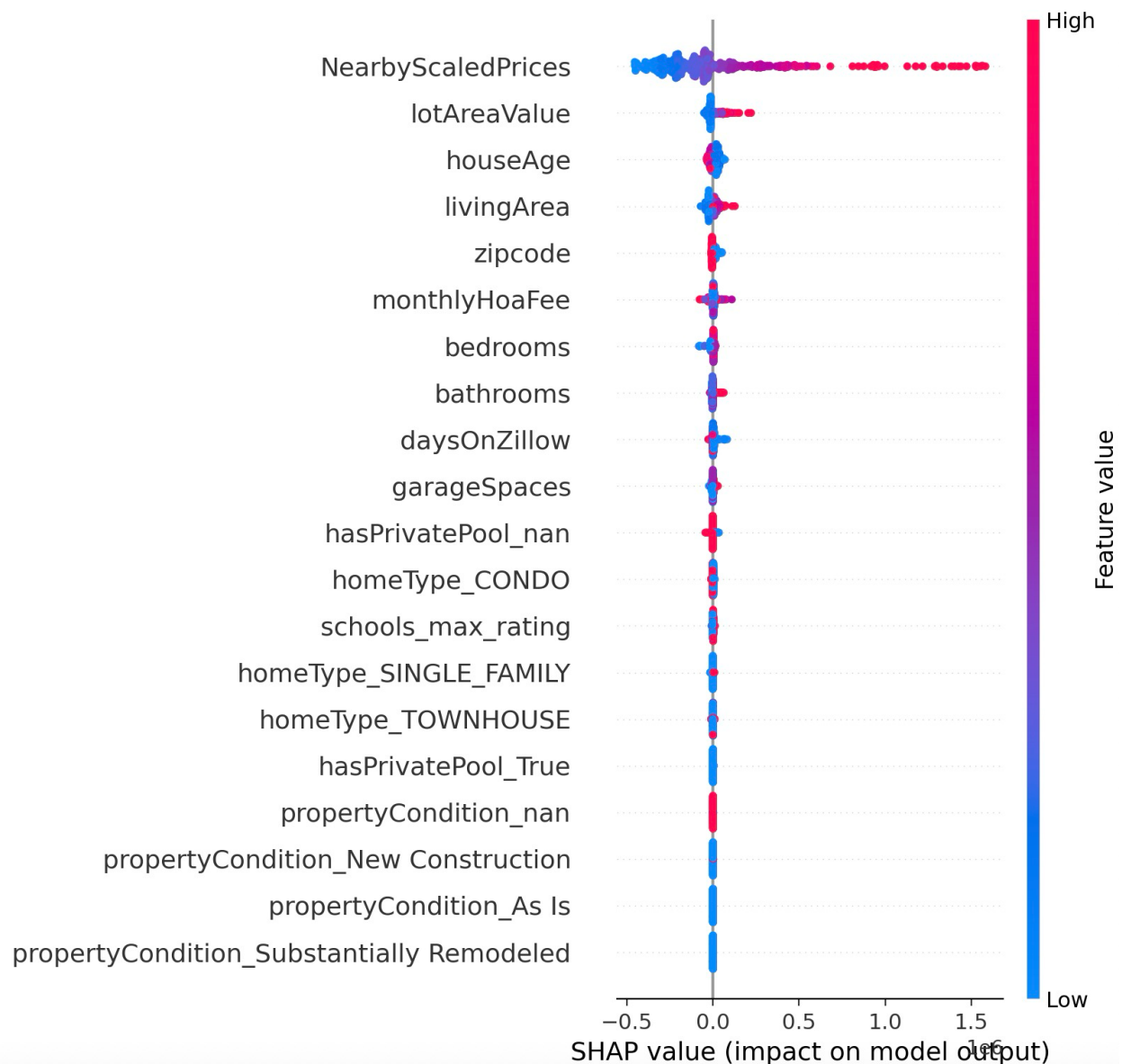
|   | Feature            | Importance | Rank |
|---|--------------------|------------|------|
| 0 | NearbyScaledPrices | 0.883547   | 1    |
| 1 | lotAreaValue       | 0.024956   | 2    |
| 2 | livingArea         | 0.024291   | 3    |
| 3 | monthlyHoaFee      | 0.019609   | 4    |
| 4 | bathrooms          | 0.012570   | 5    |
| 5 | houseAge           | 0.009599   | 6    |
| 6 | daysOnZillow       | 0.007925   | 7    |
| 7 | garageSpaces       | 0.003800   | 8    |
| 8 | hasPrivatePool_nan | 0.003478   | 9    |
| 9 | bedrooms           | 0.003341   | 10   |



The diagram illustrates a decision tree structure for classification. The root node is labeled "Age". It branches into "Young" and "Old". The "Young" branch leads to a node labeled "Gender", which branches into "Male" and "Female". The "Old" branch leads to a node labeled "Married", which branches into "Yes" and "No". Each of these branches continues to split based on other features like "Income", "Education", and "Occupation", eventually leading to leaf nodes representing specific classes.



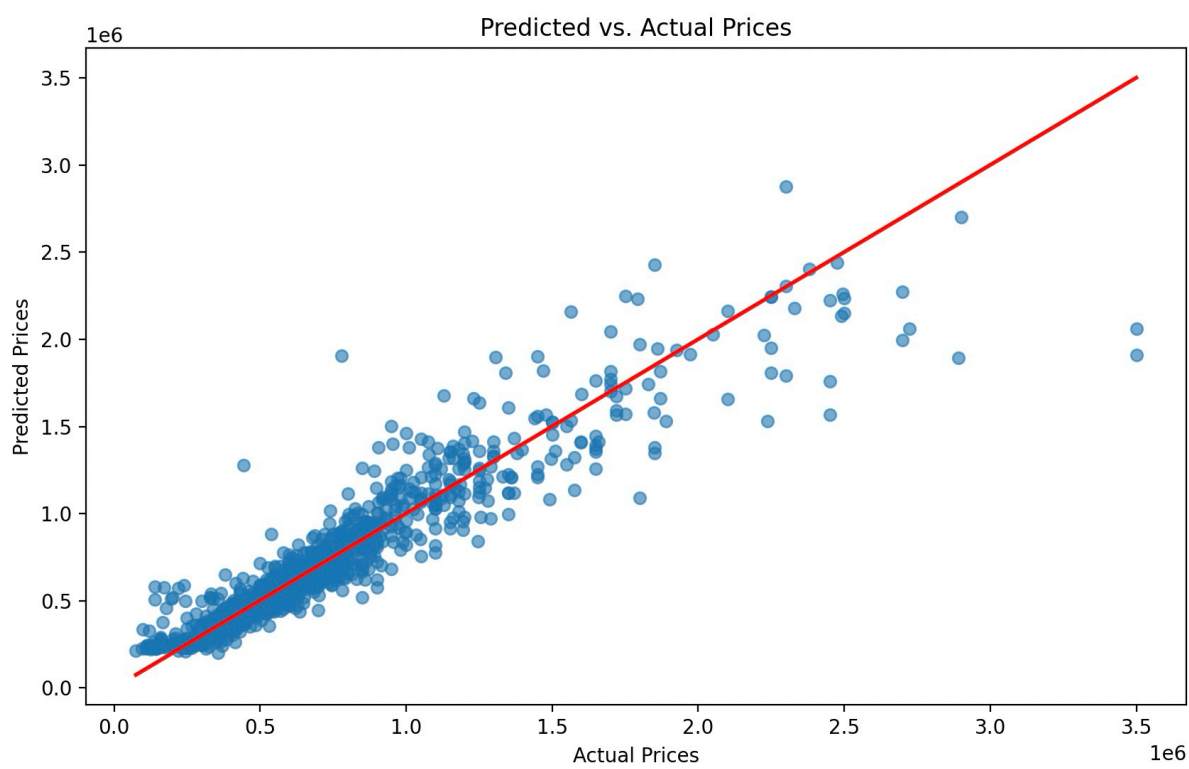
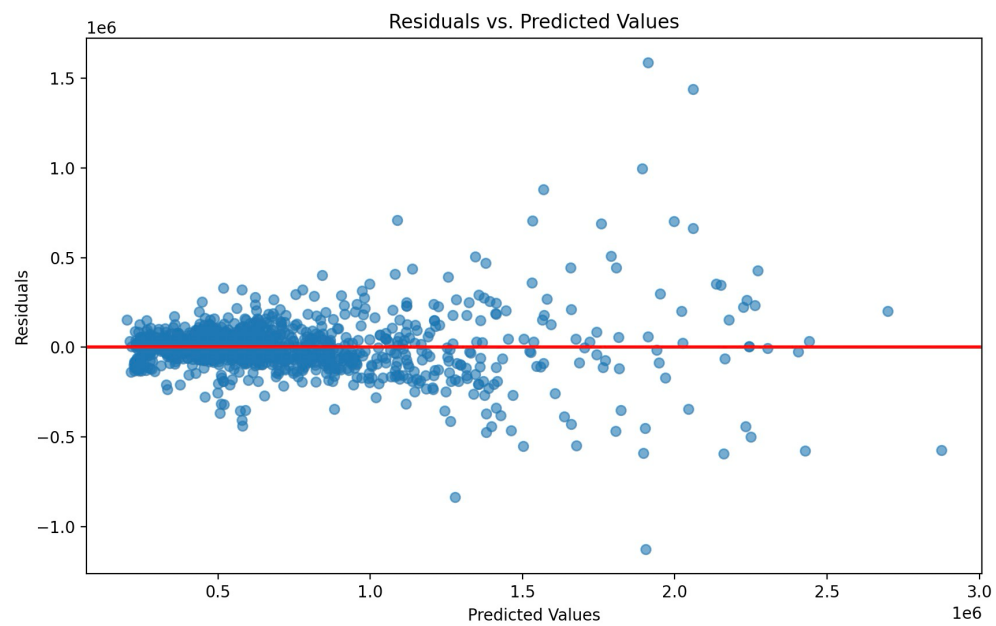


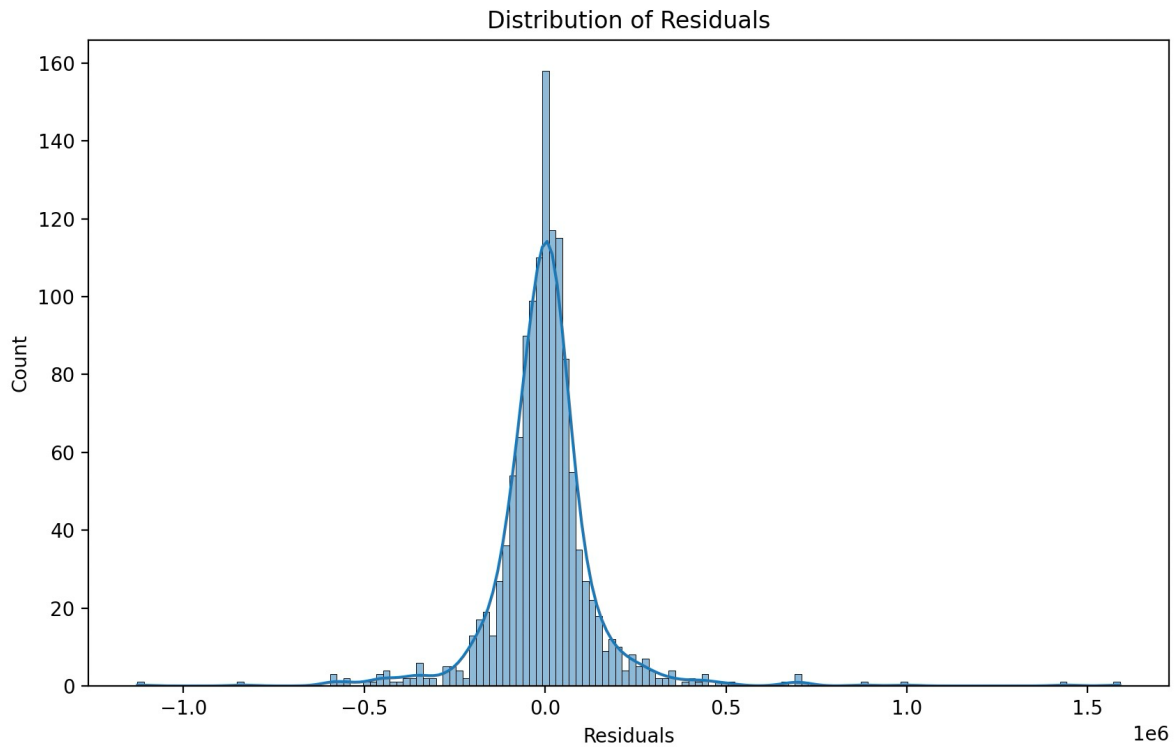


### A3: Doral, FL (Neural Network)

Cross-Validation  $R^2$  Scores: [0.8575647212739252, 0.8144358652742081, 0.9003769058304959, 0.7901376280087872, 0.810948726813759]  
Average  $R^2$  Score: 0.8347

Cross-Validation MAE Scores: [91846.31723610846, 90575.59858654869, 95270.60172021943, 92272.34467758202, 107769.19533129066]  
Average MAE Score: 95546.81





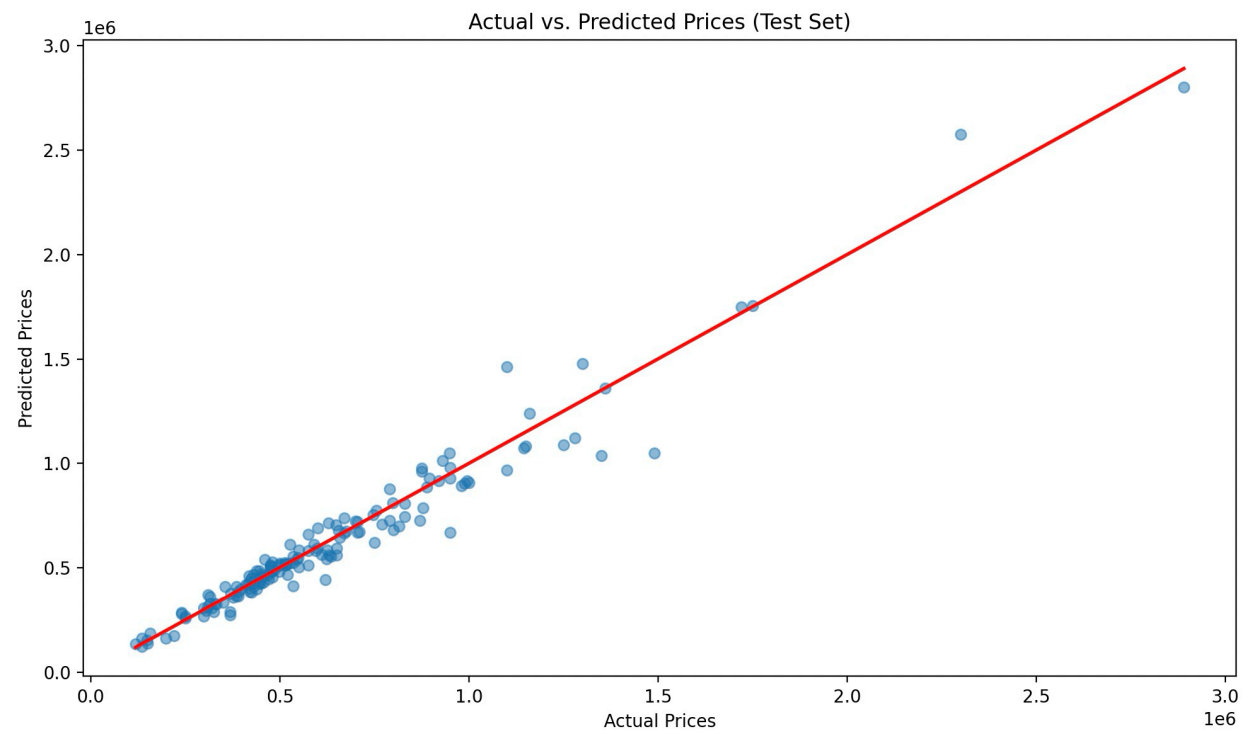
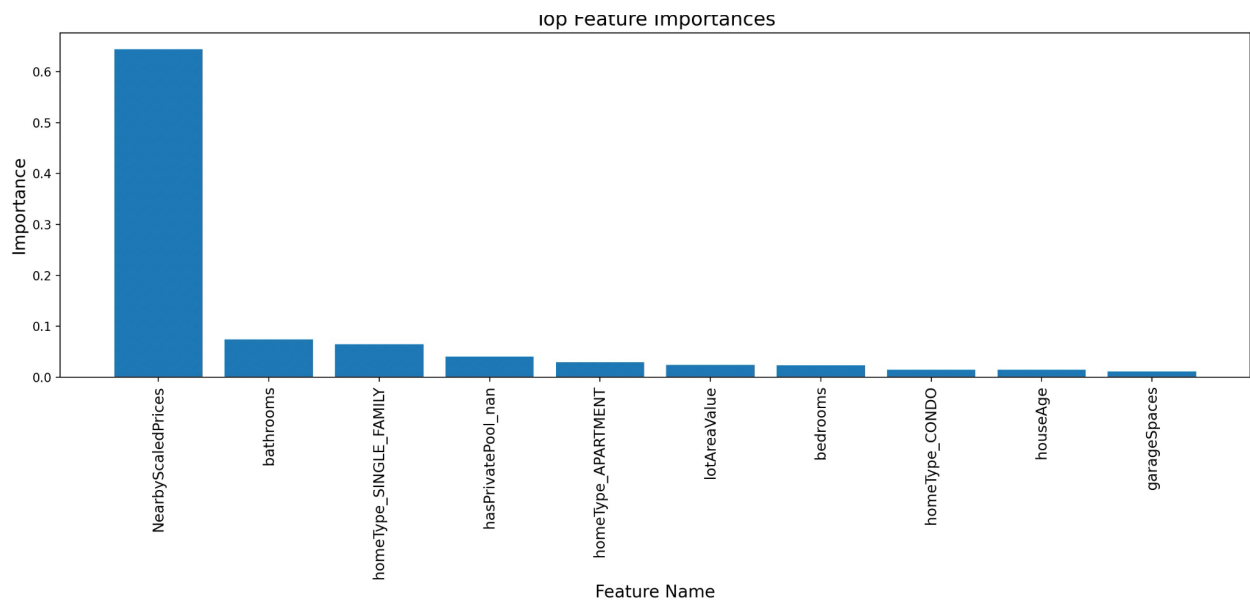
#### A4: Doral, FL (Gradient Boosting)

Cross-Validation  $R^2$  Scores: [0.9195181727409363, 0.917912483215332, 0.9536491632461548, 0.8913530707359314, 0.8236044645309448]  
Average  $R^2$  Score: 0.9012

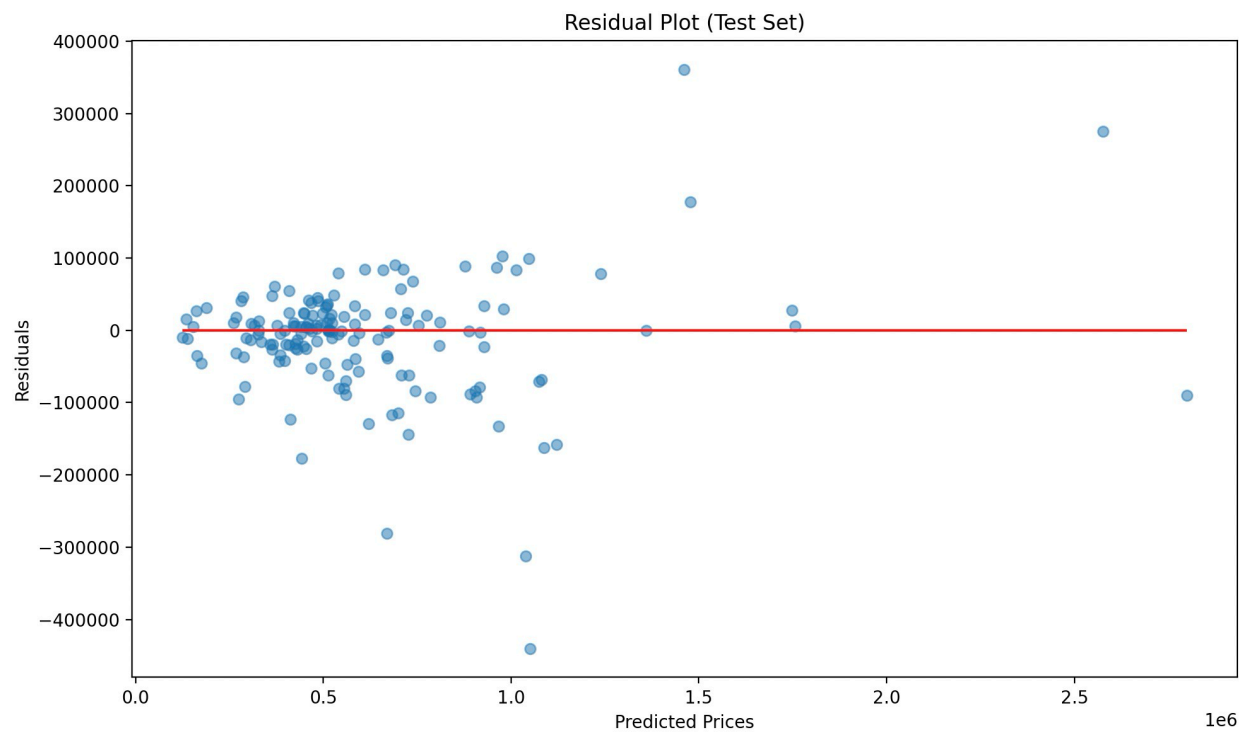
Cross-Validation MAE Scores: [68924.3203125, 65318.9375, 58612.53125, 83394.765625, 82130.9140625]  
Average MAE Score: 71676.29

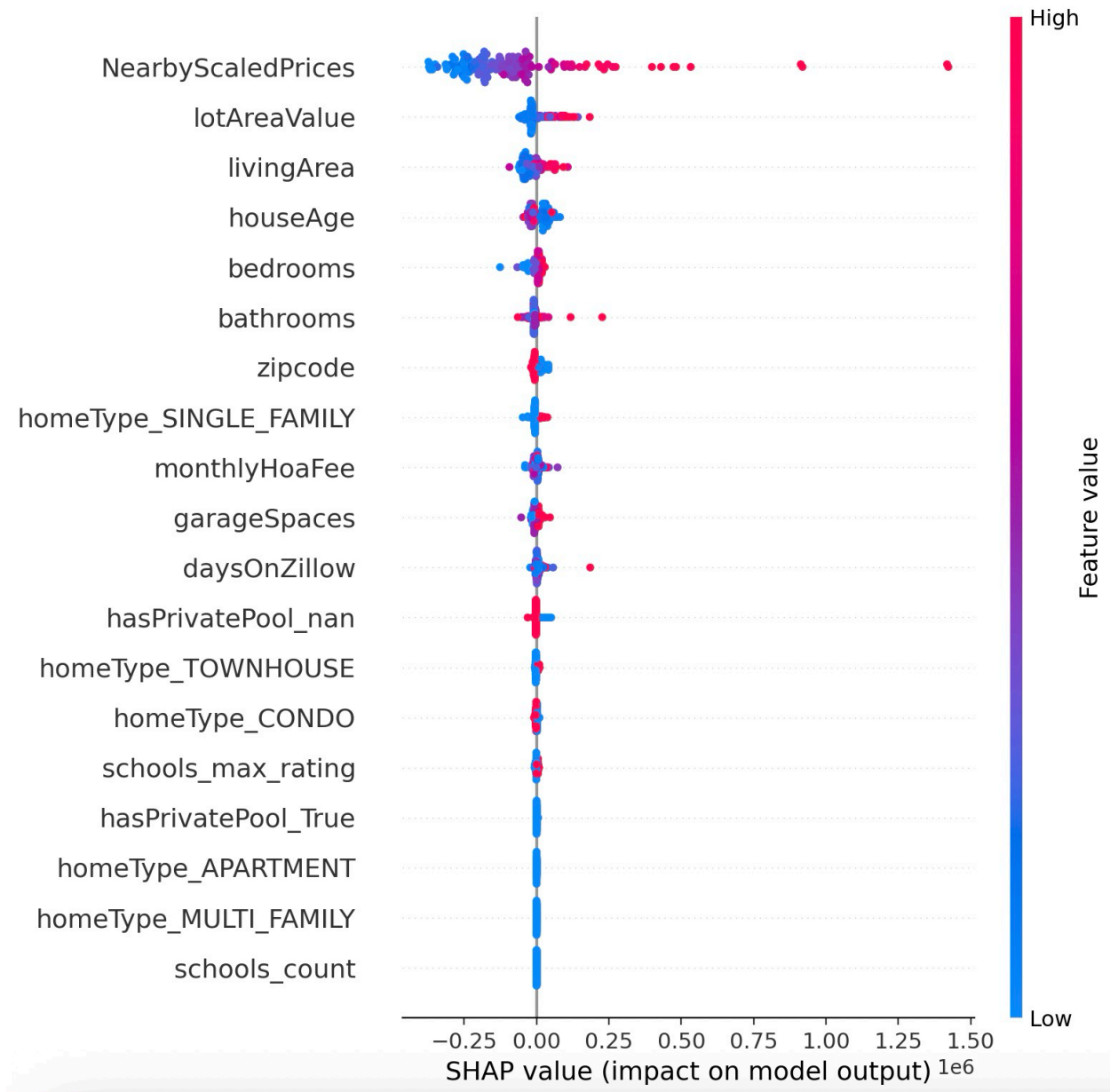
Top 10 Most Important Features:

|   | Feature                | Importance | Rank |
|---|------------------------|------------|------|
| 0 | NearbyScaledPrices     | 0.644292   | 1    |
| 1 | bathrooms              | 0.074713   | 2    |
| 2 | homeType_SINGLE_FAMILY | 0.064841   | 3    |
| 3 | hasPrivatePool_nan     | 0.040541   | 4    |
| 4 | homeType_APARTMENT     | 0.029170   | 5    |
| 5 | lotAreaValue           | 0.023819   | 6    |
| 6 | bedrooms               | 0.023233   | 7    |
| 7 | homeType_CONDO         | 0.014678   | 8    |
| 8 | houseAge               | 0.014374   | 9    |
| 9 | garageSpaces           | 0.010837   | 10   |









#### A5: Miami, FL (MLR)

Cross-Validation  $R^2$  Scores: [0.7429039574171821, 0.3454954708310979, 0.6518815740771685, 0.8889423762322356, 0.5134878165999941]  
 Average  $R^2$  Score: 0.6285

Cross-Validation MAE Scores: [978240.3178063786, 1226237.9008175188, 888121.613068051, 867409.3533985922, 1062038.1182288306]

Average MAE Score: 1004409.46

Model Summary:

WLS Regression Results

=====

Dep. Variable: y R-squared: 0.693  
Model: WLS Adj. R-squared: 0.690  
Method: Least Squares F-statistic: 195.0  
Date: Sun, 20 Apr 2025 Prob (F-statistic): 0.00  
Time: 12:15:28 Log-Likelihood: -25413.  
No. Observations: 1571 AIC: 5.086e+04  
Df Residuals: 1552 BIC: 5.097e+04  
Df Model: 18  
Covariance Type: nonrobust

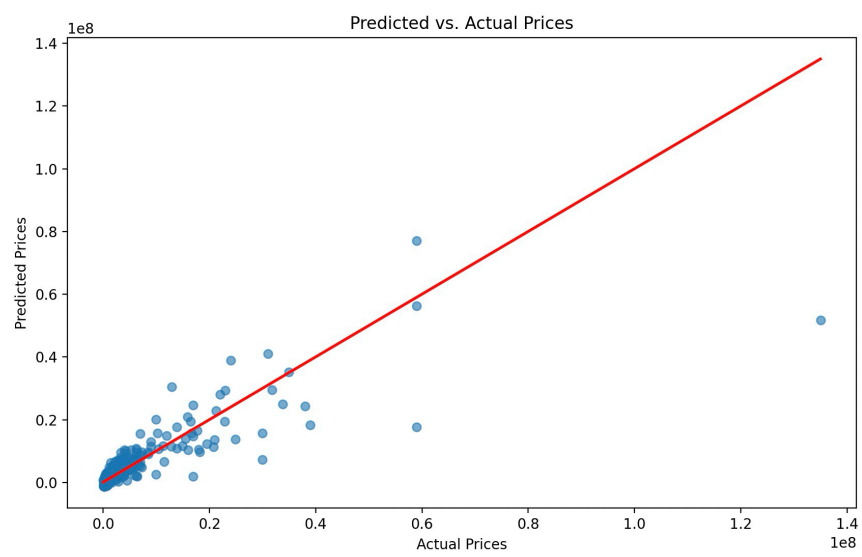
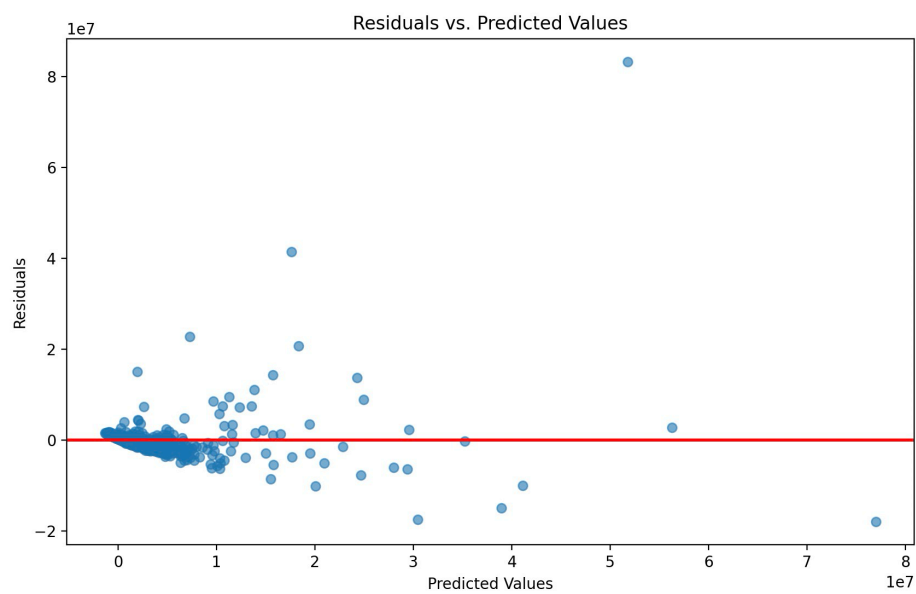
=====

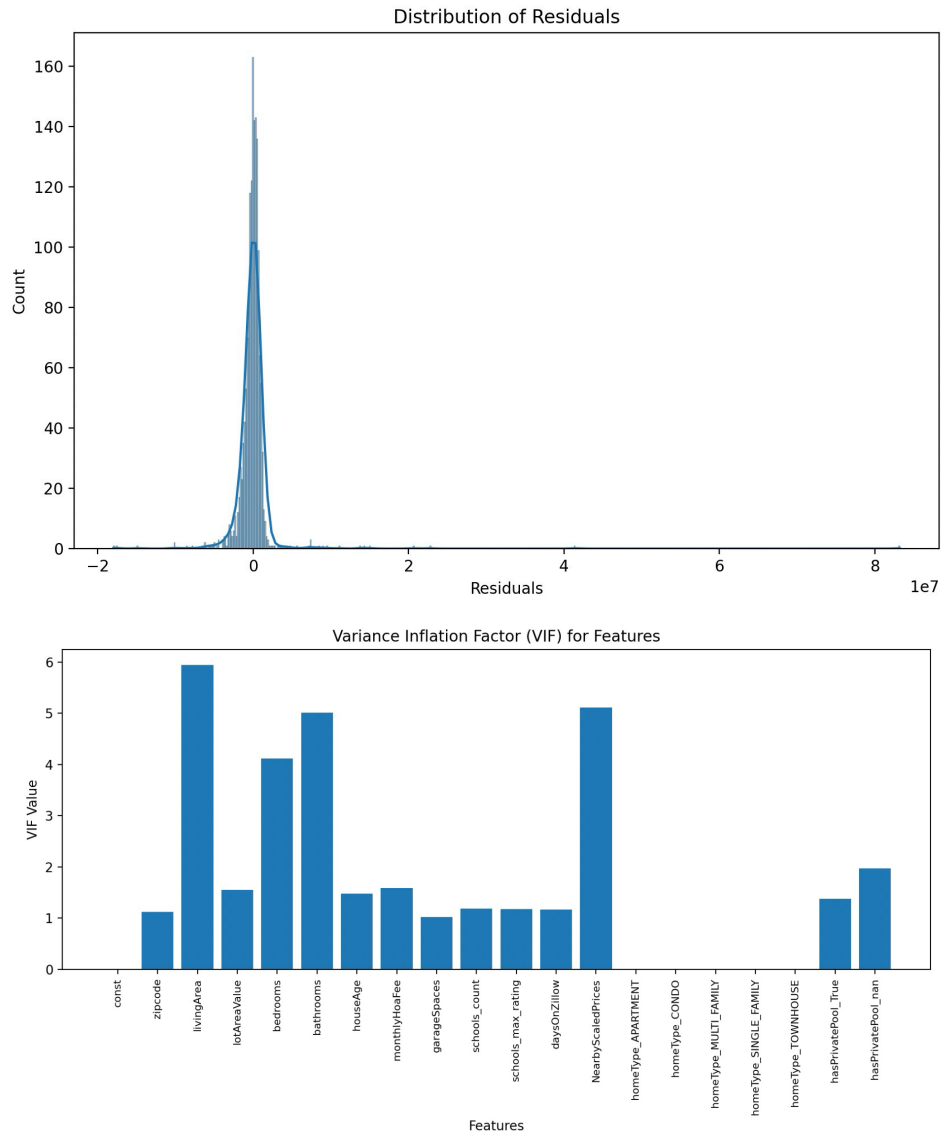
|                        | coef       | std err  | t      | P> t  | [0.025    | 0.975]    |
|------------------------|------------|----------|--------|-------|-----------|-----------|
| const                  | 9.632e+05  | 2.63e+05 | 3.661  | 0.000 | 4.47e+05  | 1.48e+06  |
| zipcode                | 3924.6402  | 3.96e+04 | 0.099  | 0.921 | -7.37e+04 | 8.15e+04  |
| livingArea             | 1.443e+05  | 9.1e+04  | 1.586  | 0.113 | -3.41e+04 | 3.23e+05  |
| lotAreaValue           | -1.861e+05 | 7.49e+04 | -2.484 | 0.013 | -3.33e+05 | -3.92e+04 |
| bedrooms               | -6.587e+04 | 1.24e+05 | -0.533 | 0.594 | -3.08e+05 | 1.77e+05  |
| bathrooms              | 4.616e+05  | 1.07e+05 | 4.326  | 0.000 | 2.52e+05  | 6.71e+05  |
| houseAge               | 2.258e+05  | 7.67e+04 | 2.946  | 0.003 | 7.54e+04  | 3.76e+05  |
| monthlyHoaFee          | 6.854e+05  | 3.94e+04 | 17.383 | 0.000 | 6.08e+05  | 7.63e+05  |
| garageSpaces           | 3.887e+04  | 7.81e+04 | 0.498  | 0.619 | -1.14e+05 | 1.92e+05  |
| schools_count          | 6.838e+04  | 3.21e+06 | 0.021  | 0.983 | -6.23e+06 | 6.37e+06  |
| schools_max_rating     | 1.278e+05  | 6.79e+04 | 1.882  | 0.060 | -5385.232 | 2.61e+05  |
| daysOnZillow           | 6.148e+04  | 1.63e+04 | 3.776  | 0.000 | 2.95e+04  | 9.34e+04  |
| NearbyScaledPrices     | 1.083e+06  | 6.76e+04 | 16.005 | 0.000 | 9.5e+05   | 1.22e+06  |
| homeType_APARTMENT     | 1.729e+05  | 8.41e+05 | 0.205  | 0.837 | -1.48e+06 | 1.82e+06  |
| homeType_CONDO         | 2.321e+05  | 2.38e+05 | 0.975  | 0.330 | -2.35e+05 | 6.99e+05  |
| homeType_MULTI_FAMILY  | -4.605e+05 | 3.6e+05  | -1.278 | 0.202 | -1.17e+06 | 2.46e+05  |
| homeType_SINGLE_FAMILY | 4.648e+05  | 2.05e+05 | 2.269  | 0.023 | 6.3e+04   | 8.67e+05  |
| homeType_TOWNHOUSE     | 5.539e+05  | 2.45e+05 | 2.262  | 0.024 | 7.37e+04  | 1.03e+06  |
| hasPrivatePool_True    | -8.475e+05 | 2.89e+05 | -2.928 | 0.003 | -1.42e+06 | -2.8e+05  |
| hasPrivatePool_nan     | -3.719e+05 | 2.35e+05 | -1.580 | 0.114 | -8.34e+05 | 8.99e+04  |

=====

Omnibus: 3109.504 Durbin-Watson: 2.077  
Prob(Omnibus): 0.000 Jarque-Bera (JB): 9863354.550  
Skew: 15.038 Prob(JB): 0.00

Kurtosis: 390.010    Cond. No. 4.16e+16

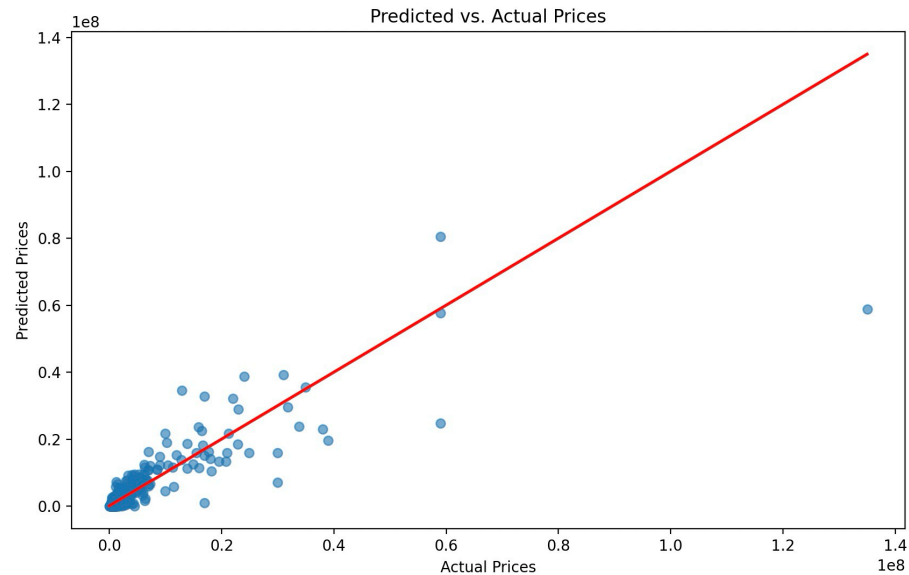
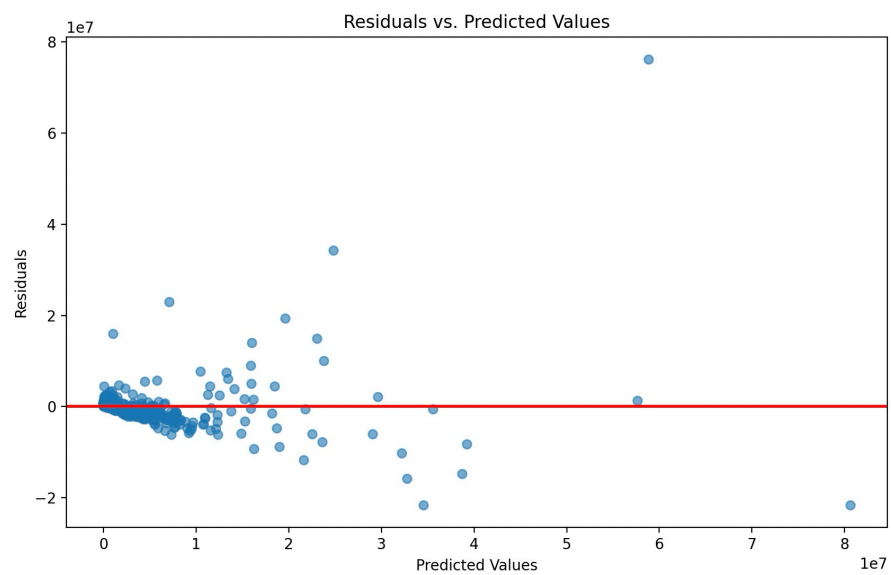


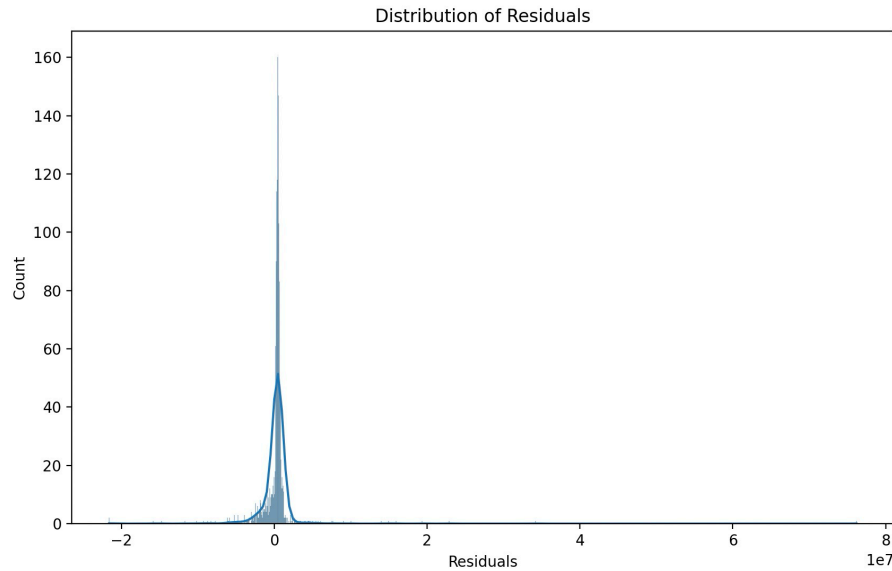


## A6: Miami, FL (Neural Network)

Cross-Validation  $R^2$  Scores: [0.7482968988720884, 0.5968064909888797, 0.6910509978679474, 0.8944911622605651, 0.5891699027394908]  
Average  $R^2$  Score: 0.7040

Cross-Validation MAE Scores: [970439.55752933, 1113045.467526885, 902836.0642009784, 825600.4810312373, 1012736.1204070795]  
Average MAE Score: 964931.54





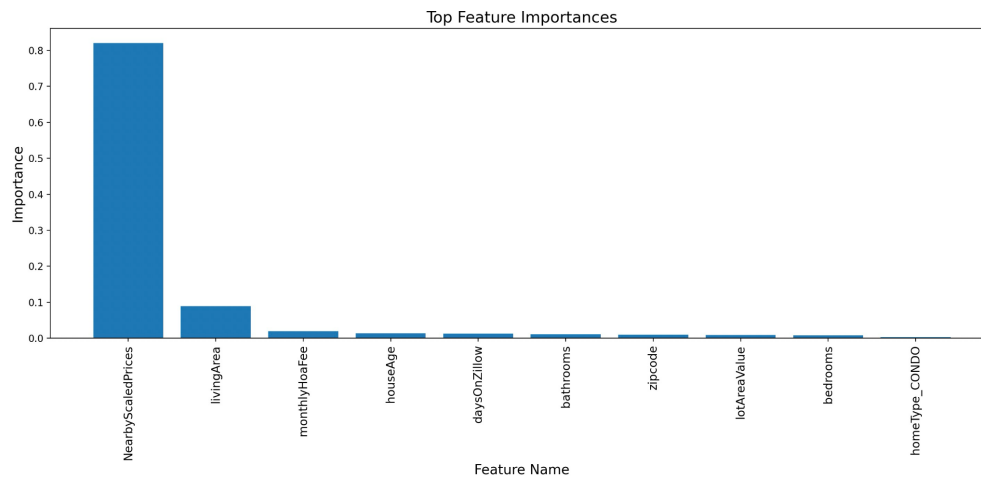
### A7: Miami, FL (Random Forest)

Cross-Validation  $R^2$  Scores: [0.7477899352438666, 0.729794618556911, 0.37131676946805015, 0.752476460365045, 0.4006364364736237]  
Average  $R^2$  Score: 0.6004

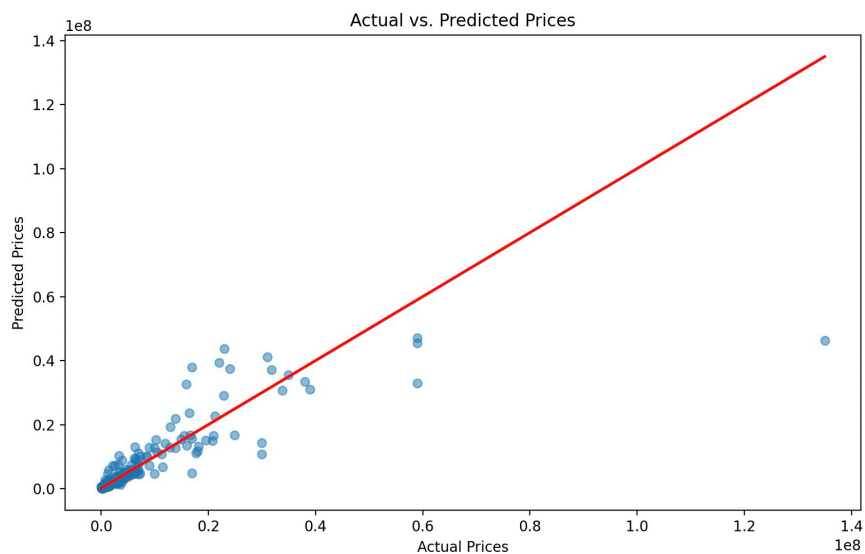
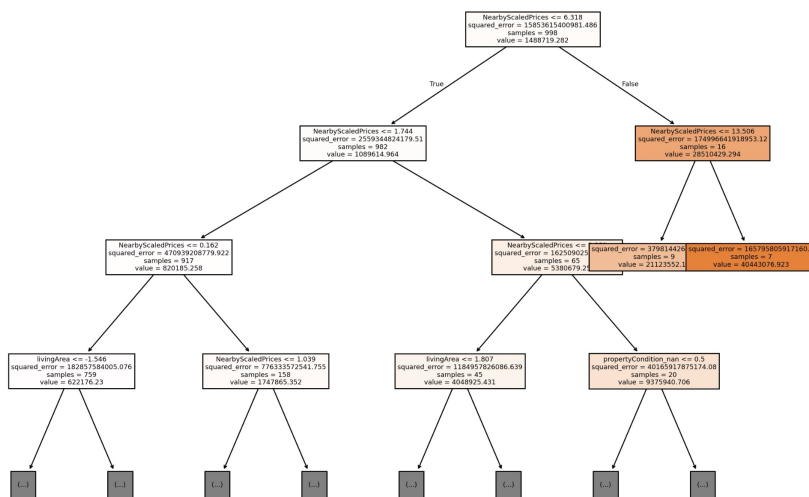
Cross-Validation MAE Scores: [696019.6934906247, 629781.9128803398, 474186.16168584896, 463112.9111330346, 851613.1719682758]  
Average MAE Score: 622942.77

Top 10 Most Important Features:

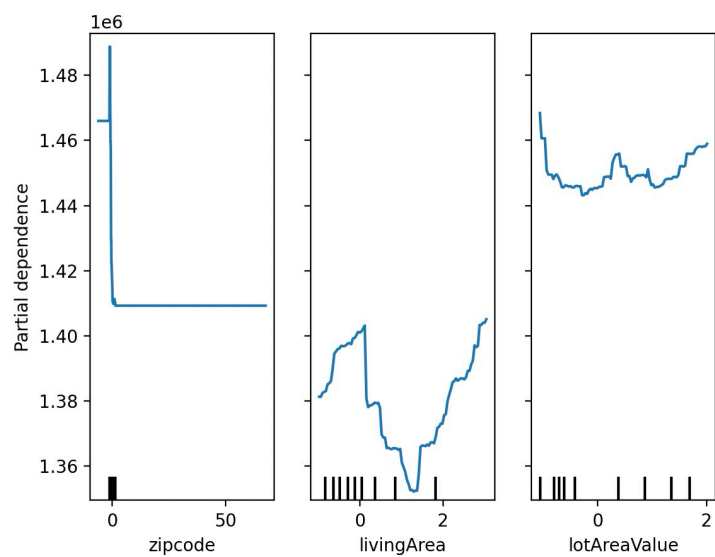
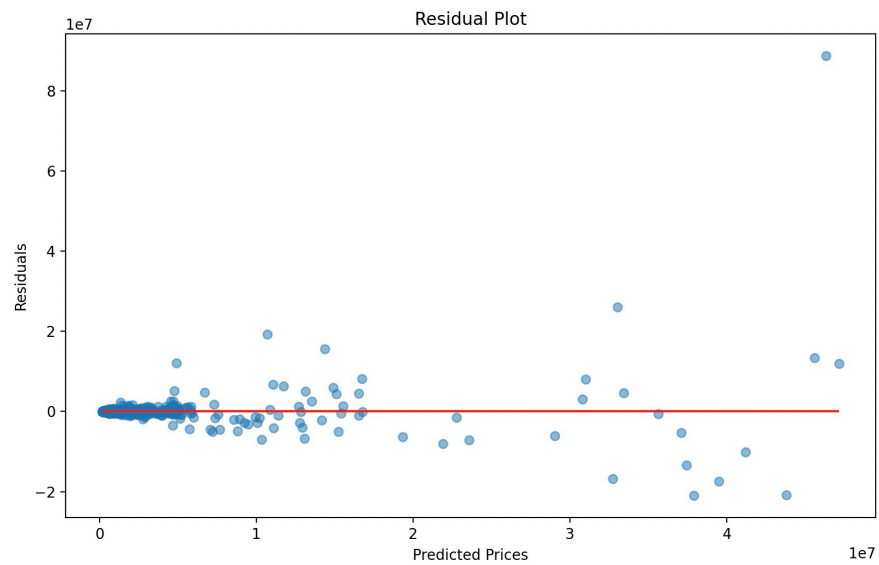
|   | Feature            | Importance | Rank |
|---|--------------------|------------|------|
| 0 | NearbyScaledPrices | 0.820364   | 1    |
| 1 | livingArea         | 0.089119   | 2    |
| 2 | monthlyHoaFee      | 0.019627   | 3    |
| 3 | houseAge           | 0.013296   | 4    |
| 4 | daysOnZillow       | 0.012308   | 5    |
| 5 | bathrooms          | 0.010725   | 6    |
| 6 | zipcode            | 0.009445   | 7    |
| 7 | lotAreaValue       | 0.009156   | 8    |
| 8 | bedrooms           | 0.007389   | 9    |
| 9 | homeType_CONDO     | 0.002945   | 10   |



Visualization of a Single Decision Tree (Depth = 3)









## A8: Miami, FL (Gradient Boosting)

Cross-Validation  $R^2$  Scores: [0.6444737315177917, 0.23387658596038818, 0.6711200475692749, 0.8375511765480042, 0.4205016493797302]

Average  $R^2$  Score: 0.5615

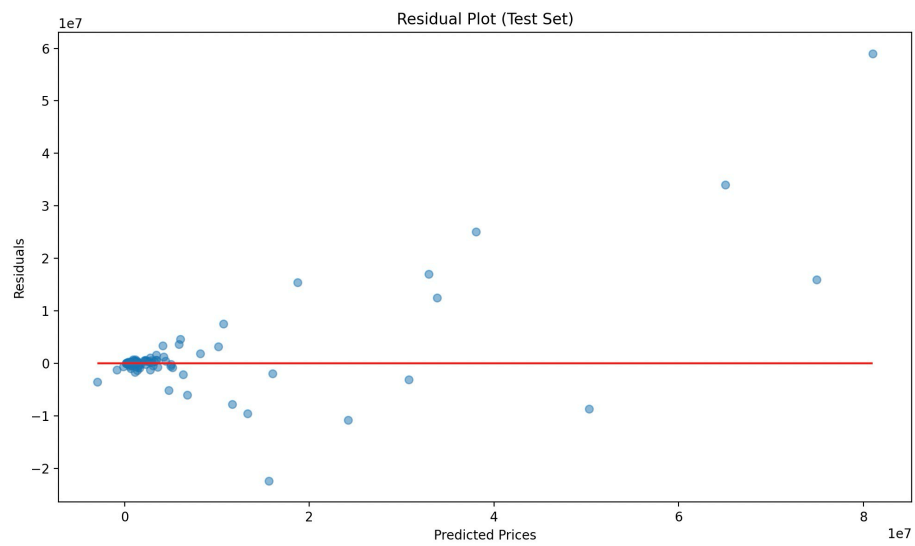
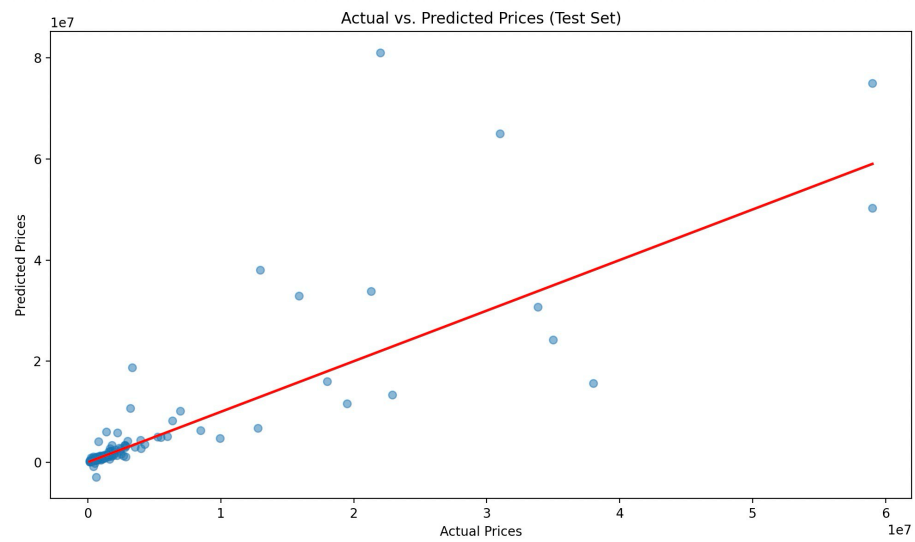
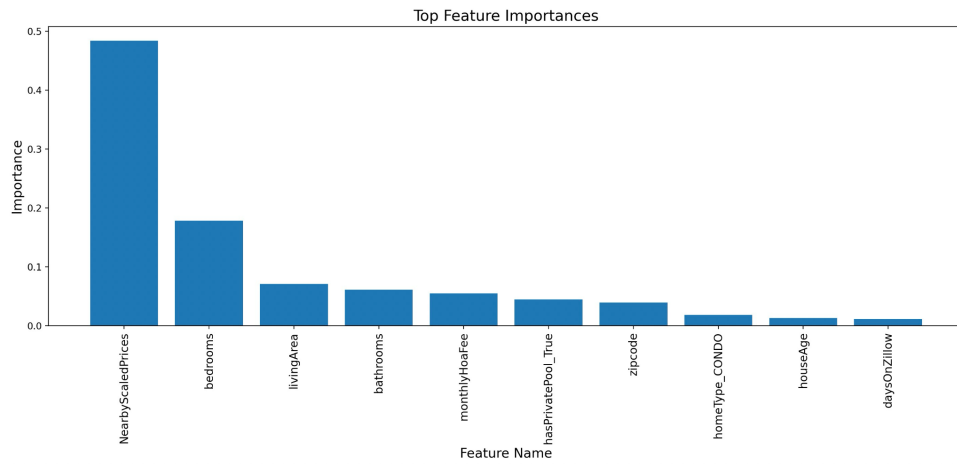
Cross-Validation MAE Scores: [475695.46875, 516228.5, 527097.4375, 373012.125, 903818.3125]

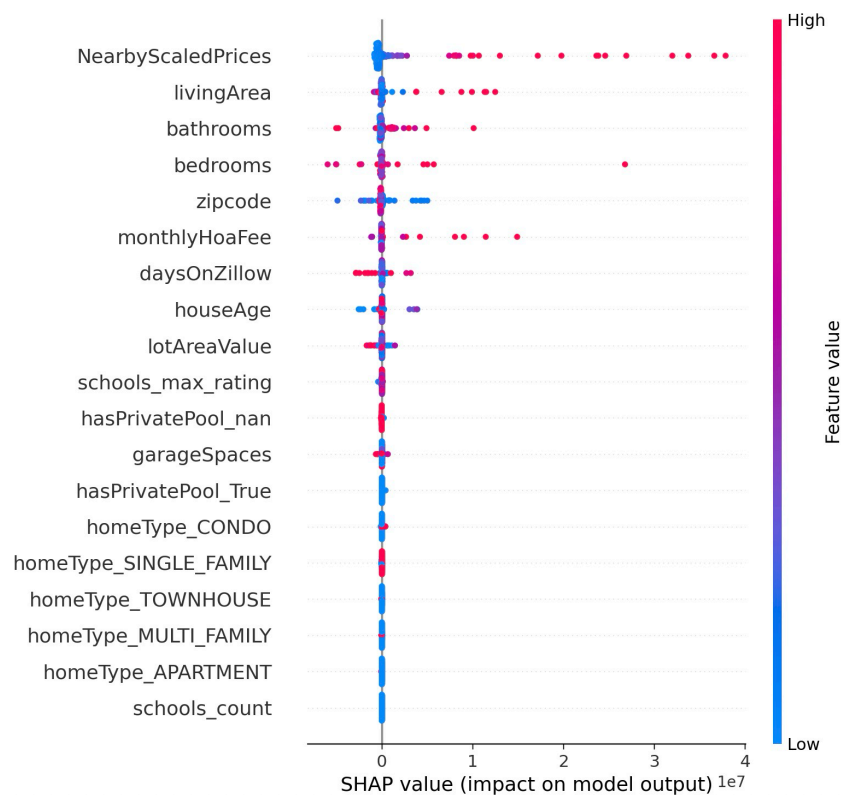
Average MAE Score: 559170.37

Top 10 Most Important Features:

|   | Feature             | Importance | Rank |
|---|---------------------|------------|------|
| 0 | NearbyScaledPrices  | 0.484009   | 1    |
| 1 | bedrooms            | 0.177975   | 2    |
| 2 | livingArea          | 0.070399   | 3    |
| 3 | bathrooms           | 0.060682   | 4    |
| 4 | monthlyHoaFee       | 0.054630   | 5    |
| 5 | hasPrivatePool_True | 0.044593   | 6    |
| 6 | zipcode             | 0.039016   | 7    |
| 7 | homeType_CONDO      | 0.018014   | 8    |
| 8 | houseAge            | 0.013111   | 9    |

9      daysOnZillow    0.011472    10





## A9: Kendall, Florida (MLR)

Cross-Validation  $R^2$  Scores: [0.8873920039972001, 0.9145948187324914, 0.8778189329974209, 0.8984617780431501, 0.8491576640992229]

Average  $R^2$  Score: 0.8855

Cross-Validation MAE Scores: [148485.10630255035, 128787.46739520972, 150680.18305634664, 144105.55858288833, 152048.11651061574]

Average MAE Score: 144821.29

Model Summary:

### WLS Regression Results

```
=====
=====
Dep. Variable:          y   R-squared:                0.895
Model:                  WLS   Adj. R-squared:          0.893
Method:                 Least Squares   F-statistic:      581.4
Date:                   Sun, 20 Apr 2025   Prob (F-statistic): 0.00
Time:                   12:35:54   Log-Likelihood:    -16232.
No. Observations:       1178   AIC:                3.250e+04
Df Residuals:           1160   BIC:                3.259e+04
Df Model:               17
```

Covariance Type: nonrobust

=====

|                        | coef       | std err  | t       | P> t  | [0.025    | 0.975]    |
|------------------------|------------|----------|---------|-------|-----------|-----------|
| const                  | 9.206e+05  | 3.63e+04 | 25.346  | 0.000 | 8.49e+05  | 9.92e+05  |
| zipcode                | -2.748e+04 | 8286.427 | -3.316  | 0.001 | -4.37e+04 | -1.12e+04 |
| livingArea             | 1.163e+05  | 2.51e+04 | 4.642   | 0.000 | 6.72e+04  | 1.66e+05  |
| lotAreaValue           | -4.205e+04 | 7505.958 | -5.602  | 0.000 | -5.68e+04 | -2.73e+04 |
| bedrooms               | -1.437e+04 | 1.54e+04 | -0.936  | 0.350 | -4.45e+04 | 1.58e+04  |
| bathrooms              | 1.017e+05  | 1.28e+04 | 7.931   | 0.000 | 7.65e+04  | 1.27e+05  |
| houseAge               | -1.878e+04 | 8352.596 | -2.248  | 0.025 | -3.52e+04 | -2387.318 |
| monthlyHoaFee          | -2.082e+04 | 8023.759 | -2.595  | 0.010 | -3.66e+04 | -5077.588 |
| garageSpaces           | 3.125e+04  | 8645.151 | 3.615   | 0.000 | 1.43e+04  | 4.82e+04  |
| schools_count          | 1.639e-10  | 9.96e-11 | 1.645   | 0.100 | -3.15e-11 | 3.59e-10  |
| schools_max_rating     | 1.614e+04  | 7739.749 | 2.086   | 0.037 | 958.961   | 3.13e+04  |
| daysOnZillow           | -1710.6754 | 6528.623 | -0.262  | 0.793 | -1.45e+04 | 1.11e+04  |
| NearbyScaledPrices     | 3.437e+05  | 2.2e+04  | 15.610  | 0.000 | 3.01e+05  | 3.87e+05  |
| homeType_APARTMENT     | 1.298e+05  | 6.28e+04 | 2.068   | 0.039 | 6629.144  | 2.53e+05  |
| homeType_CONDO         | 1.723e+05  | 2.72e+04 | 6.328   | 0.000 | 1.19e+05  | 2.26e+05  |
| homeType_MULTI_FAMILY  | 1.825e+05  | 1.02e+05 | 1.785   | 0.075 | -1.81e+04 | 3.83e+05  |
| homeType_SINGLE_FAMILY | 2.834e+05  | 2.99e+04 | 9.492   | 0.000 | 2.25e+05  | 3.42e+05  |
| homeType_TOWNHOUSE     | 1.526e+05  | 2.82e+04 | 5.405   | 0.000 | 9.72e+04  | 2.08e+05  |
| hasPrivatePool_True    | -3.262e+05 | 3.4e+04  | -9.597  | 0.000 | -3.93e+05 | -2.59e+05 |
| hasPrivatePool_nan     | -4.176e+05 | 3.54e+04 | -11.802 | 0.000 | -4.87e+05 | -3.48e+05 |

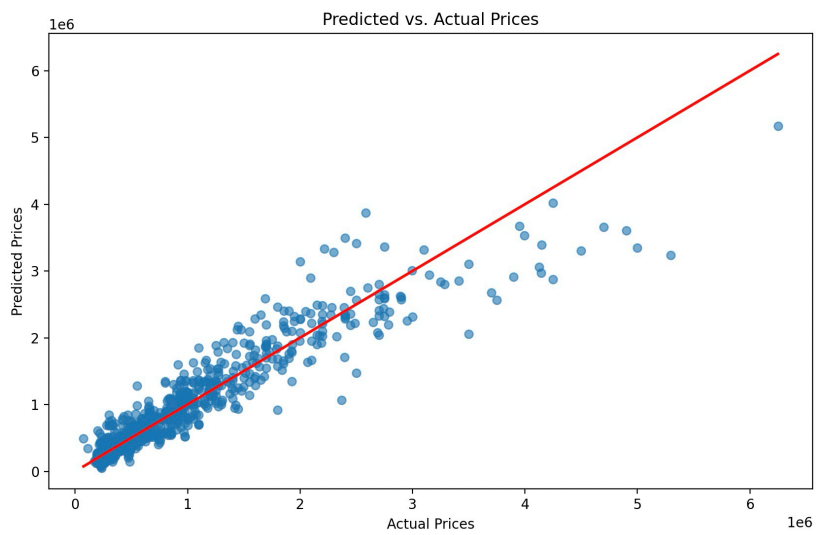
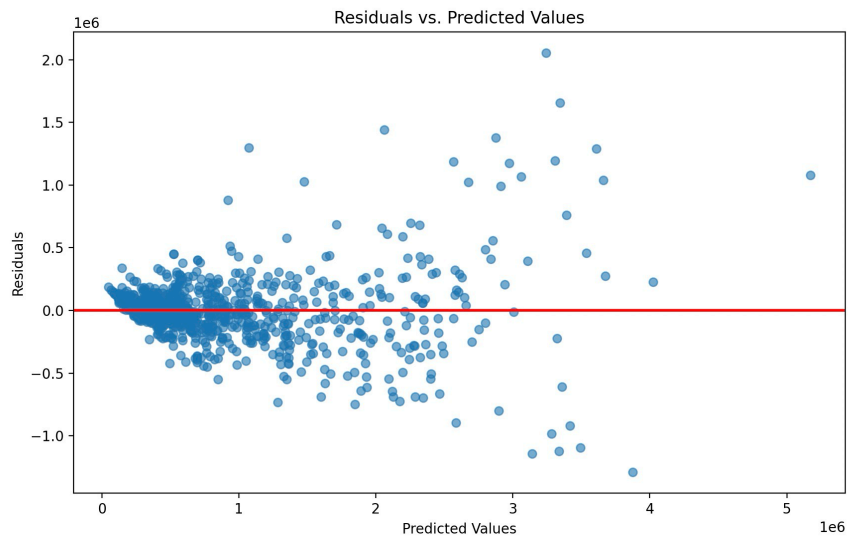
=====

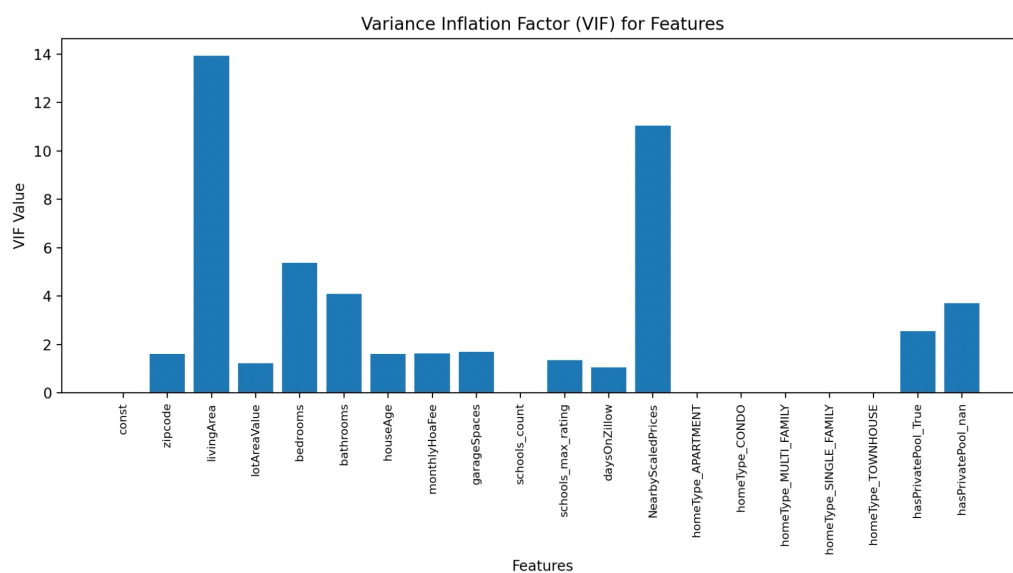
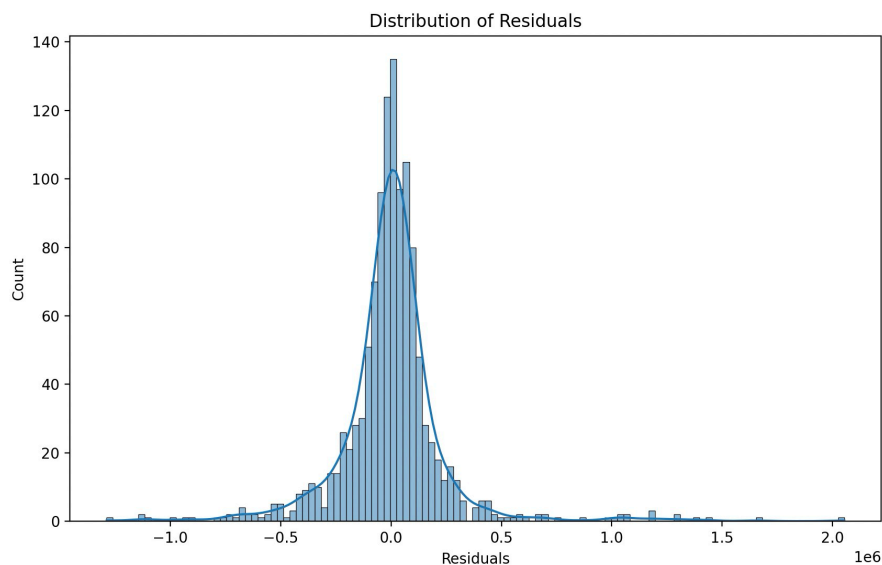
=====

|                |         |                   |          |
|----------------|---------|-------------------|----------|
| Omnibus:       | 390.476 | Durbin-Watson:    | 2.079    |
| Prob(Omnibus): | 0.000   | Jarque-Bera (JB): | 5084.146 |
| Skew:          | 1.152   | Prob(JB):         | 0.00     |
| Kurtosis:      | 12.913  | Cond. No.         | 4.94e+16 |

=====

=====

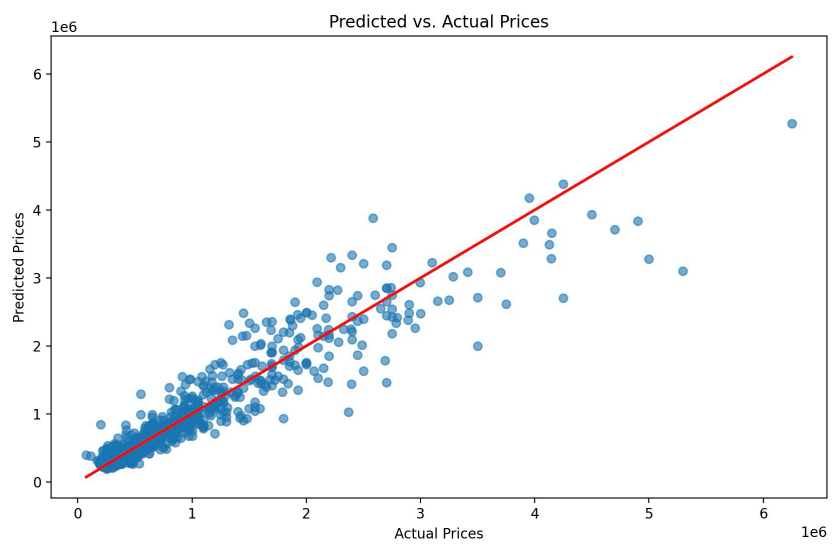
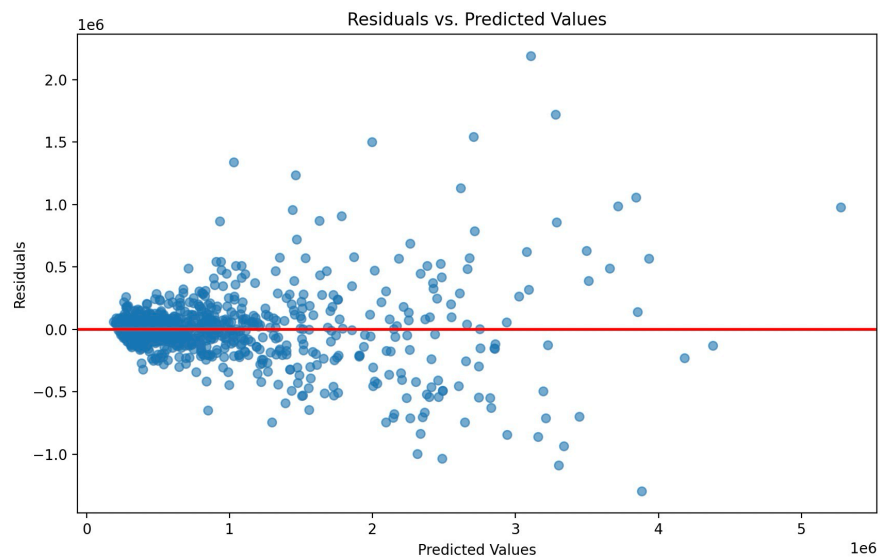




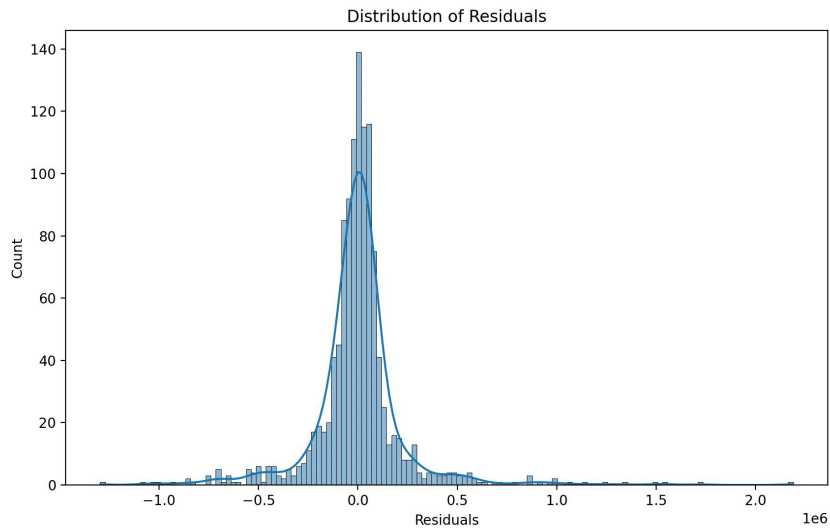
## A10: Kendall, FL (Neural Network)

Cross-Validation  $R^2$  Scores: [0.8673435131538443, 0.9057774241173253, 0.8563822487185468, 0.9264211462318559, 0.8462367113361048]  
 Average  $R^2$  Score: 0.8804

Cross-Validation MAE Scores: [151310.8996111123, 129953.68633460795, 156523.8268504505, 125735.13740227673, 162630.95101749816]  
 Average MAE Score: 145230.90







### A11: Kendall, FL (Random Forest)

Cross-Validation  $R^2$  Scores: [0.8871918112923475, 0.8856031011092511, 0.879078083977598, 0.915528476323735, 0.8841440155587404]

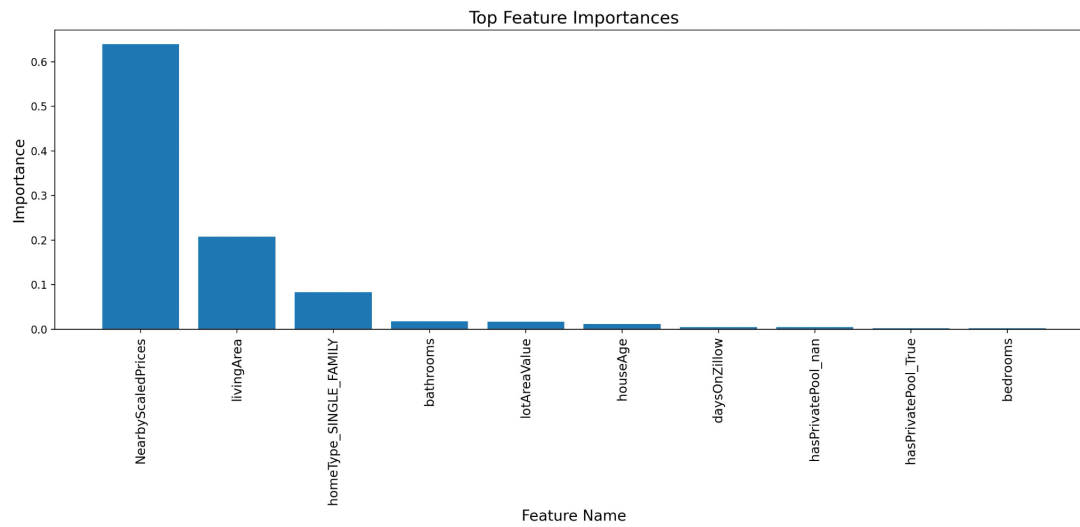
Average  $R^2$  Score: 0.8903

Cross-Validation MAE Scores: [133328.86358587095, 107662.49710093047, 127456.92065900637, 107847.02265920436, 126118.3029889784]

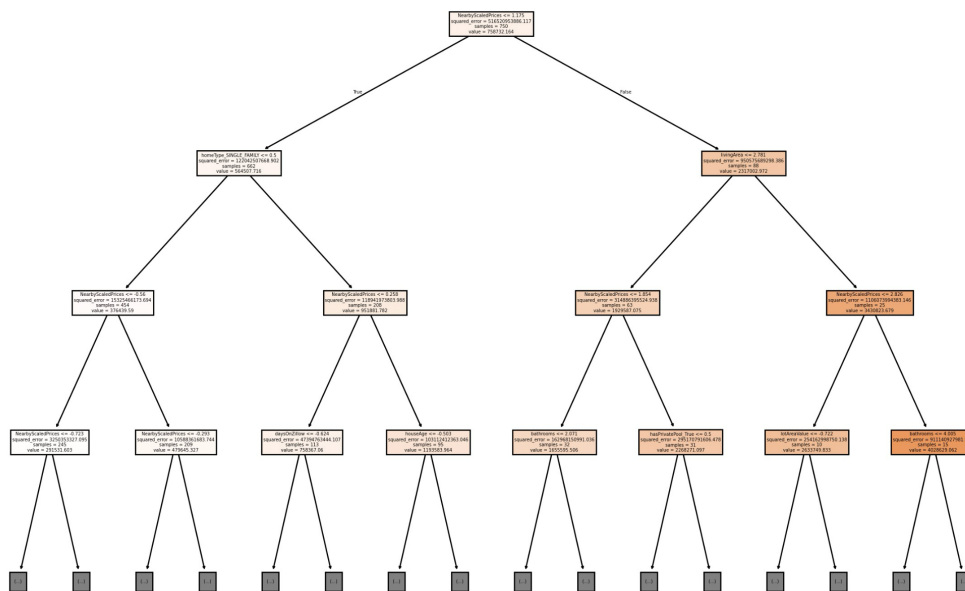
Average MAE Score: 120482.72

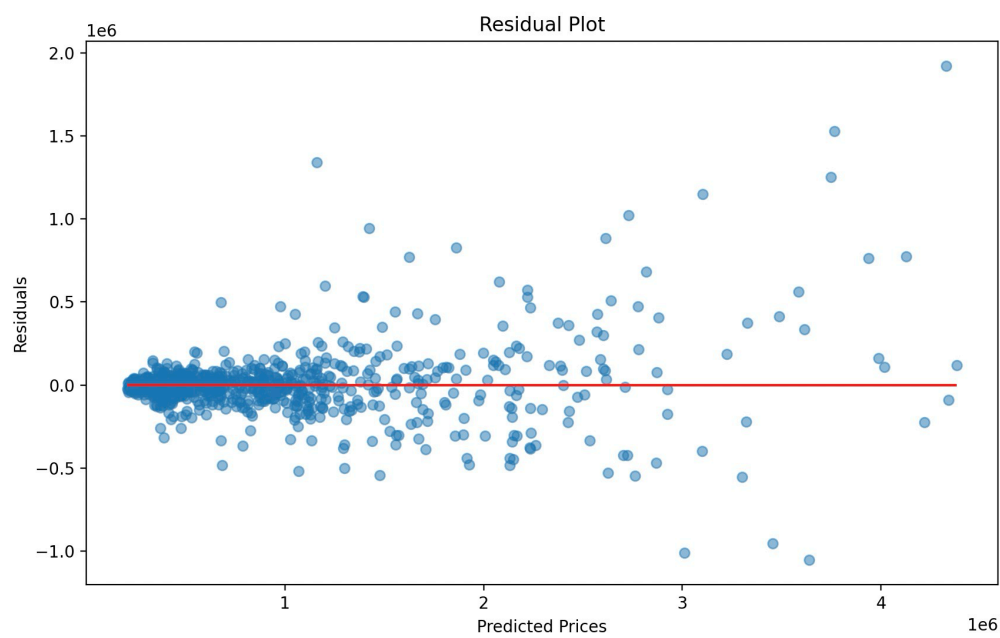
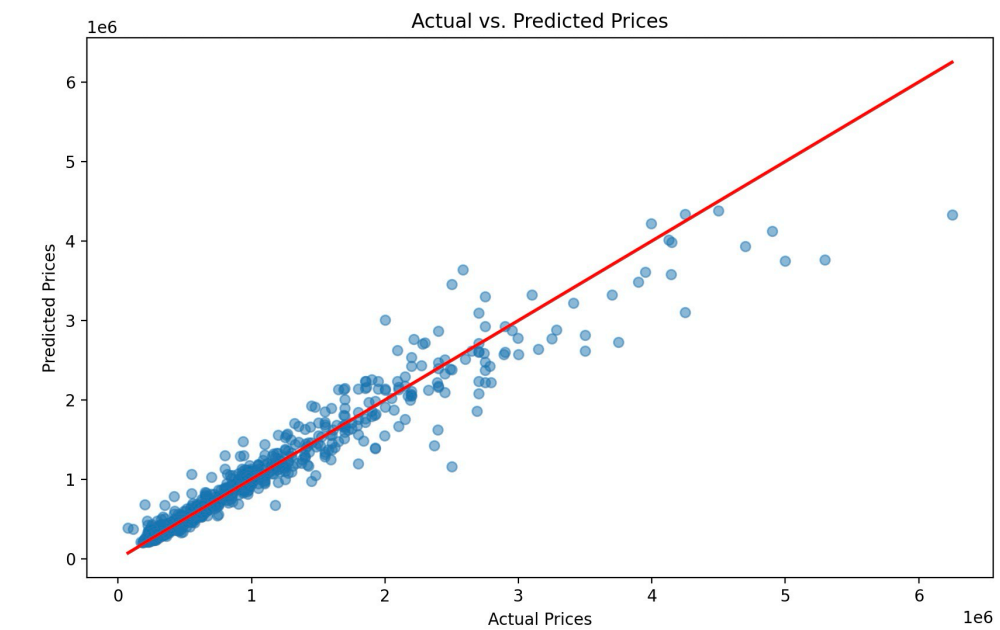
Top 10 Most Important Features:

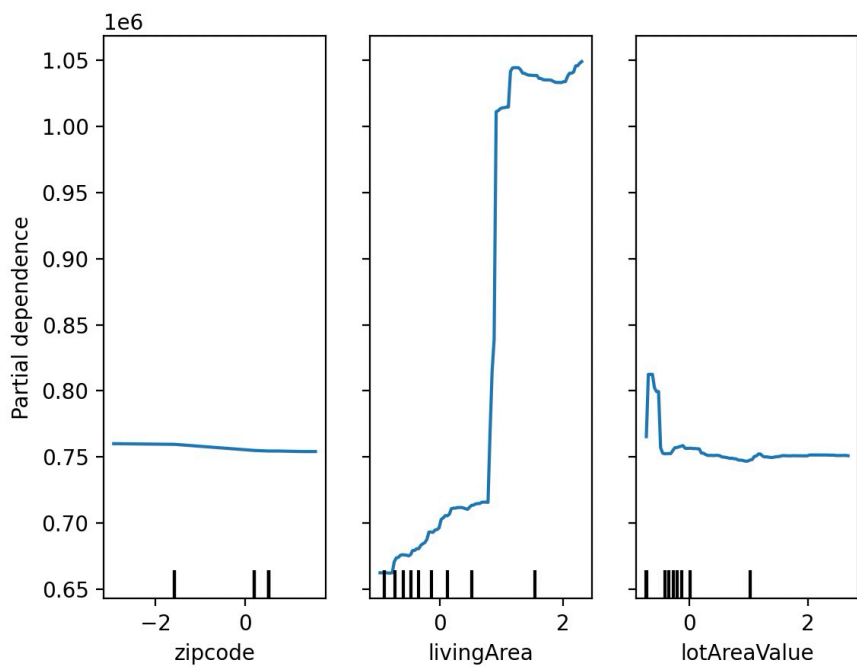
|   | Feature                | Importance | Rank |
|---|------------------------|------------|------|
| 0 | NearbyScaledPrices     | 0.639174   | 1    |
| 1 | livingArea             | 0.207777   | 2    |
| 2 | homeType_SINGLE_FAMILY | 0.083384   | 3    |
| 3 | bathrooms              | 0.018071   | 4    |
| 4 | lotAreaValue           | 0.016779   | 5    |
| 5 | houseAge               | 0.012162   | 6    |
| 6 | daysOnZillow           | 0.005299   | 7    |
| 7 | hasPrivatePool_nan     | 0.004862   | 8    |
| 8 | hasPrivatePool_True    | 0.002670   | 9    |
| 9 | bedrooms               | 0.002639   | 10   |

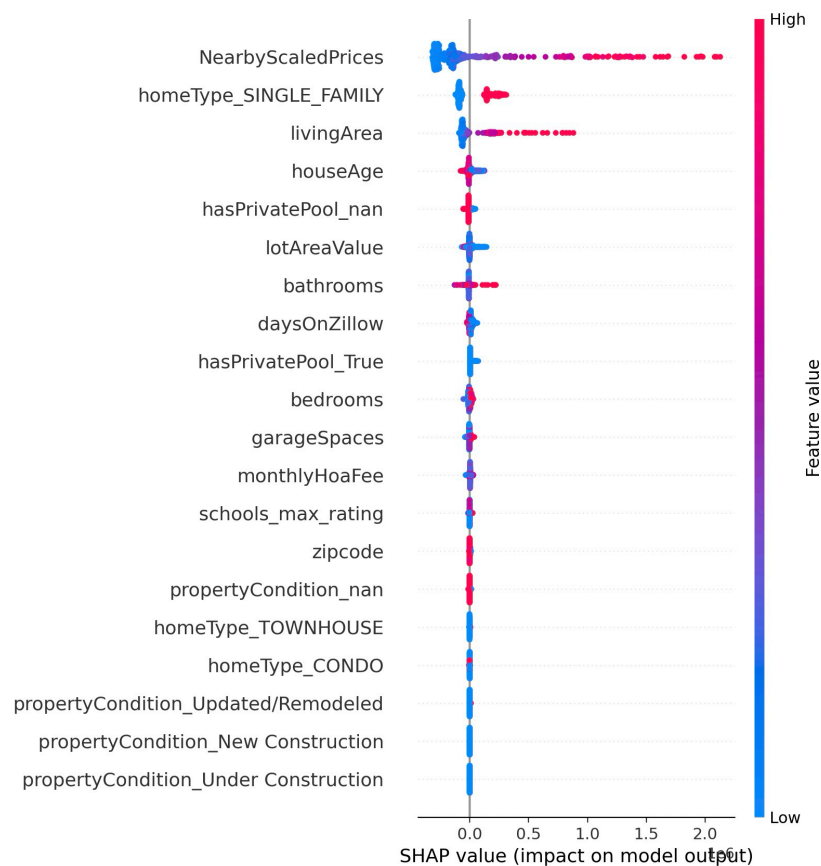


Visualization of a Single Decision Tree (Depth = 3)









## A12: Kendall, FL (Gradient Boosting)

Cross-Validation  $R^2$  Scores: [0.9071012735366821, 0.8503992557525635, 0.8796961307525635, 0.9304397702217102, 0.8924140930175781]

Average  $R^2$  Score: 0.8920

Cross-Validation MAE Scores: [132083.171875, 117168.1484375, 99630.6015625, 122244.6796875, 107762.2109375]

Average MAE Score: 115777.76

Top 10 Most Important Features:

|   | Feature                | Importance | Rank |
|---|------------------------|------------|------|
| 0 | homeType_SINGLE_FAMILY | 0.895647   | 1    |
| 1 | NearbyScaledPrices     | 0.070577   | 2    |
| 2 | hasPrivatePool_nan     | 0.014069   | 3    |
| 3 | bathrooms              | 0.005752   | 4    |
| 4 | livingArea             | 0.003487   | 5    |
| 5 | lotAreaValue           | 0.002109   | 6    |
| 6 | bedrooms               | 0.002068   | 7    |
| 7 | hasPrivatePool_True    | 0.001966   | 8    |
| 8 | houseAge               | 0.001099   | 9    |

9    homeType\_TOWNHOUSE    0.001057    10

